Pollution Pricing in Equilibrium: Production, Reallocation, and Aggregate Impacts

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Abstract

We develop a framework to quantify the equilibrium effects of pollution pricing through three margins: short-run input substitution, reallocation across firms, and clean technology adoption. In the European Union Emissions Trading System (2005–2021) doubling the carbon price cuts contemporaneous emissions by 23%, primarily through reallocation to cleaner firms with negligible input substitution within firms. Output falls 3.6%, mostly from carbon costs passing down supply chains raising costs for downstream firms. Clean technology adoption responds endogenously to prices, but with a lag. Over five years it deepens the emissions cut to 31% and eases the output decline to 2.5%.

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1 Introduction

Pricing pollution is the textbook fix for an externality. But how do these policies achieve emissions reductions, and at what cost? The answer depends on how firms and markets adjust in equilibrium when pollution is priced. Firms may adjust production processes, make longer-run investments to reduce emissions, while market forces may shift production across firms.

Quantifying these adjustments has remained difficult, even as pollution pricing has proliferated.[1] Two obstacles stand in the way. First, abatement effort is usually unobserved: standard firm data show emissions and input use, but not how inputs are split between production and abatement. Second, even once firm-level responses are estimated, they must still be aggregated to evaluate a policy's total effect. Equilibrium forces drive reallocation across firms and propagate cost shocks through the production network, shaping aggregate and distributional outcomes. We develop a microfounded model of firm production and abatement that overcomes each of these obstacles. We show how to estimate the model from standard firm data without directly observing abatement effort, and how the model translates firm-level responses into aggregate equilibrium outcomes.

Our framework captures three adjustment margins through which pollution pricing reduces aggregate emissions. The first is firms' short-run *input substitution*: firms may change which materials they purchase or which types of workers they hire in order to cut emissions, for example, by shifting electricity production within the firm from coal to gas boilers already in place. The second is *reallocation* of output from dirty to cleaner firms: pollution pricing increases output prices of dirty firms relative to clean firms, so consumers substitute toward less emissions-intensive producers. The third is firms' long-run *clean technology adoption*: firms respond to higher pollution prices by investing in clean, emissions-saving technology, such as building new gas plants to retire existing coal plants. Clean technology accumulates over years due to time-to-build frictions, gradually reducing the emissions intensity of production.

At the core of our model are firms' production and abatement choices. In each period, given their existing production technology, firms combine capital, labor, and materials to produce output, generating emissions as a byproduct that can be reduced through costly abatement. Abatement effort is unobserved in most empirical settings.[2] We show that under weak assumptions about the relationship between abatement and production, the firm's problem is equivalent to one in which emissions enter directly as an input alongside capital, labor, and materials. This reformulation generalizes the Cobb-Douglas structures used in recent work to a broad class of production technologies (Copeland and Taylor, 2004; Shapiro and Walker, 2018; Hollingsworth et al., 2026). In this representation, the elasticity of substitution between emissions and other inputs *is* the ease of abatement: when it is small, emissions and other inputs are near-complements, abatement is costly, and firms cannot cut emissions without cutting output.

[1] Carbon pricing is now implemented in over 50 countries, covering nearly a third of global carbon emissions (World Bank, 2026).

[2] In our setting using common firm financial data, we observe aggregate expenditures on labor, capital, and materials, but not whether any given input is emitting or whether any given input is allocated toward abatement.

In our model, monopolistically competitive firms sell output both to consumers for final consumption and to other firms for use as inputs. This opens two equilibrium channels for aggregate abatement. First, the reallocation margin can operate directly by shifting final demand and production toward less emissions-intensive producers—both within and across sectors. Demand shifts toward cleaner producers can reduce aggregate emissions even if no individual firm's emissions intensity changes. Second, because many regulated, highly polluting sectors are central in the production network, their higher emissions costs pass through to other downstream firms and sectors, which can induce another set of reallocation and aggregate emissions effects. Our approach to estimating these effects follows the work of Oberfield and Raval (2021).

While a firm's production technology is fixed within a period, over time firms invest in clean technology. In our model, clean technology is a persistent emissions-saving productivity term that lets firms produce the same output with fewer emissions, or analogously, their inputs directed toward abatement become more productive. Because clean investments take years to plan, build, and bring online, this margin adjusts gradually.[3]

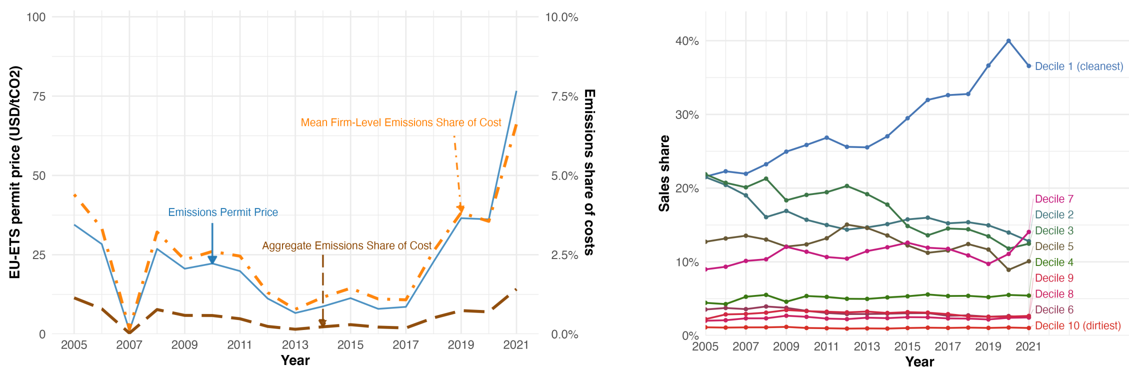
We use our model to study the first 17 years of the European Union Emissions Trading System (EU-ETS), the world's first and largest carbon market. Since its introduction in 2005, EU-ETS has regulated emissions from power generation, manufacturing, and other energy-intensive sectors through a cap-and-trade system. Over our sample period, aggregate emissions at regulated installations declined by 35%, while industrial production remained relatively stable. These aggregate trends motivate the core policy questions we address: how much does carbon pricing reduce emissions, what mechanisms drive these reductions, what are the short- and long-run costs of these reductions in the aggregate and across sectors?

Figure 1 presents several patterns in the data that guide our modeling approach, and previews some of our findings. In Panel (a), the evolution of EU-ETS permit prices is plotted alongside emissions cost shares—defined as a firm's emissions costs (the economic cost of retired emissions permits) divided by total costs. There are three key takeaways from this first panel. First, year-to-year, firm-level emissions cost shares move in lockstep with the permit price. If firms could substitute away from emissions cheaply, emissions would fall relative to other inputs as its price rose, and the emissions cost share would climb more slowly than the price. Through the lens of our model, the tight comovement implies that short-run abatement is costly: firms can still cut emissions, but only by cutting output, not by shifting their inputs toward abatement. Second, over the full 2005–2021 sample, permit prices rose by roughly 120% while average firm-level emissions cost shares rose by only approximately 50%, much less than one-for-one. The longer-run decoupling of prices and firm-level emissions cost shares suggests firms reduced their emissions intensity gradually over time. Third, the economy-wide aggregate emissions cost share only weakly tracks permit prices, indicating that the market as a whole responds to higher prices by reallocating production toward less emissions-intensive firms.[4] Panel (b) substantiates this last point by showing how the sales

[3]For example, Colmer et al. (2024) find that French firms regulated under EU-ETS upgraded their capital stock, lowering emissions intensity over the several years following the introduction of the cap-and-trade program.

[4]Note that over the full sample, the aggregate emissions share of costs changes by only 0.3pp, compared to a 2.2pp

Figure 1: Permit Prices, Emissions Cost Shares, and Sales Shares by Emissions Intensity



(a) Permit Prices and Emissions Cost Shares

(b) Sales Share by Emissions Intensity

Notes: Panel (a): The solid blue line shows the average annual permit price. The dash-dotted orange line depicts the mean firm-level emissions cost share—a firm's emissions multiplied by the permit price, divided by its total input cost (the sum of capital, labor, materials, and emissions costs). The long-dashed red line shows the aggregate emissions cost share, calculated using total emissions costs and total input costs across all firms. Panel (b): the sales share of the firms in each decile of the 2005 emissions intensity distribution. Both panels use a balanced panel of firms.

share of the firms that fall into each decile of the 2005 emissions intensity distribution evolves over time. The cleanest decile of firms (i.e., decile 1) has an increasing sales share, while the other deciles have shrinking or stable sales shares.[5]

While our theoretical framework can accommodate a broad class of production technologies, our quantitative model must capture the three margins reflected in these trends: limited short run abatement, long run within-firm decarbonization, and reallocation from dirty to clean firms. We show that a nested constant elasticity of substitution (CES) specification fits the production data well, and we adopt this functional form for our baseline analysis. The inner nest combines capital, labor, and materials into a productive composite while the outer nest combines that composite with emissions. Clean technology enters the outer nest as an emissions-saving productivity term. The outer nest substitution elasticity governs the ease of abatement in the short run via input substitution. Clean technology determines the emissions that result from a given choice of productive inputs. In our model, firms take clean technology as given in any period, and endogenously invest to improve clean technology in the long run. We specify the demand side of our model as nested CES across firms within a sector, and across sectors. Nested CES allows for dirty firms or dirty sectors as a whole to shrink in response to carbon pricing while cleaner firms and sectors may grow.

Our estimates of the model parameters and counterfactual simulations reveal four key findings. First, the input substitution margin is rigid: firm-level elasticities of substitution between emissions and the productive composite are essentially zero for all sectors. Holding technology fixed, firms

increase in the average firm-level emissions share of costs.

[5]The patterns shown in Panel (b) are robust to whether we use different years to define the emissions intensity deciles. The patterns in both panels are robust to whether we instead use the full sample of firms in our EUTL-Orbis matched dataset.

cannot reduce emissions in the short run without cutting output. We identify this elasticity from the firm's cost-minimizing first-order conditions using permit supply shocks captured in the carbon price surprise series of Känzig (2025), and extended by Jézéquel-Royer and Levieuge (2025).

Second, reallocation across firms is a major driver of the aggregate emissions response to carbon pricing. Its strength depends on two parameters. The first is an emissions heterogeneity index, which captures within-sector dispersion in clean technology and how much a common carbon price disperses unit costs and output prices across firms.[6] The second is a sector-specific demand elasticity, which governs how aggressively market shares shift toward cleaner firms as those relative prices change. Both vary widely across sectors: the heterogeneity index is an order of magnitude larger in power than in paper products, and demand elasticities range from about 4 in rubber and plastics to 12 in refining. The short-run elasticity of aggregate emissions to the carbon price is -0.34, but collapses to -0.05 if reallocation is shut down, confirming that reallocation drives most of the short-run response. The largest reductions are concentrated in power, cement, and refining: the first two through substantial heterogeneity in clean technology, refining through highly elastic demand.

Third, clean technology responds endogenously to carbon prices, but with a lag. The average firm's clean technology improved by approximately 46% over our sample period (Appendix Figure A8), but this observed trend reflects many drivers—fossil fuel prices, exogenous cost reductions in clean technology over time, and other regulatory mandates as well as carbon pricing. To isolate the causal contribution of the carbon price, we estimate the dynamic response of clean technology to permit price shocks via local projections, using the same carbon price surprise series as the source of identifying variation in carbon prices. We find that increases in the carbon price induce firms to shift investment toward cleaner technologies: pooling across firms, the five-year permit price elasticity of clean technology is approximately 0.211—meaning a 1% permit price increase raises firms' clean technology by roughly 0.211% within five years—increasing to 0.318 when sectors are weighted by their share of emissions, because the largest clean technology responses are in carbon-intensive sectors such as refining and power generation.

Fourth, our counterfactuals show carbon pricing moves emissions several times more than it moves overall economic activity. Doubling the permit price cuts aggregate EU-ETS emissions by 23% in the short run while output falls by only 3.6%. After five years of induced clean technology adoption, the emissions cut deepens to 31%[7] while easing the output loss to 2.5%. The asymmetry has a clear source. Reallocation of output toward cleaner firms in direct response to emissions pricing affords deep emissions reductions with moderate aggregate output declines. The output declines that do occur are almost entirely because upstream carbon costs pass down the production network to downstream firms, even clean ones.[8] For example, a higher carbon price raises electricity

[6] In general, the strength of reallocation depends on heterogeneity in firms' emissions cost shares, which are also a function of other production parameters. Since we find that emissions are statistically perfect complements with productive inputs, the cost share is solely a function of clean technology and factor prices.

[7] This is less than the 39% a naïve sum of the two margins would suggest, because the 16pp contributed by clean technology adoption on its own partly substitute for reallocation.

[8] We also find limited leakage to non-ETS firms. Because non-ETS emissions are unobserved, we impute them

prices which increases input costs for downstream manufacturing firms, thus reducing their output.

These effects vary sharply across sectors. Economic costs are highest where demand is inelastic (cement, power) or where exposure to upstream costs passed-through the production network is greatest (metals, chemicals). Sectors less exposed through the production network, such as paper, cut emissions with less effect on output or prices. Together, the three margins reconcile a rigid short-run response at the firm level with large aggregate emissions reductions and moderate impacts on output and employment.

Our results resolve a tension between two strands of the carbon pricing literature that have proceeded in parallel. A large applied microeconomic literature uses quasi-experimental designs to identify the firm-level effects of carbon pricing on outcomes such as emissions, employment, and innovation (Petrick and Wagner, 2014; Calel and Dechezleprêtre, 2016; Marin et al., 2018; Calel, 2020; Dechezleprêtre et al., 2023; Colmer et al., 2024). Studies of firm output generally find small or null impacts (Martin et al., 2016; Dechezleprêtre et al., 2023; Colmer et al., 2024). A more recent macroeconomic literature uses time series methods to estimate the aggregate effects of carbon pricing (Metcalf and Stock, 2023; Bjørnland et al., 2024; Känzig, 2025; Barrutiabengoa Ortubai et al., 2025). Känzig (2025), in particular, finds that carbon price shocks generate sizable output contractions. Our model shows the two are consistent because they measure different objects. The direct effect of a carbon price on a firm's own output is modest. Aggregate output falls 6 times more in equilibrium because upstream carbon costs—from sectors like power—pass down the production network to downstream firms, raising their materials costs and further suppressing output. The micro estimates, by comparing regulated and unregulated firms, difference out much of this economy-wide component (Nakamura and Steinsson, 2018), precisely what the macro estimates capture. Thus, our framework builds on both literatures by bringing firm-level data to quantify aggregate equilibrium responses, which allows us to shed light on the causes of aggregate effects and the firm-level primitives that underpin them.

In addition to carbon pricing, our framework contributes to two other literatures. First, we build on a recent line of work that models emissions as an input to production in quantitative equilibrium models (Shapiro and Walker, 2018; Hollingsworth et al., 2026).[9] We derive closed-form marginal abatement cost curves at the firm, sector, and economy-wide levels that account for equilibrium reallocation, extending the aggregation methodology of Raval (2019) and Oberfield and Raval (2021) from capital-labor substitution to emissions abatement. Second, by identifying the causal response of clean technology adoption to carbon pricing, we contribute to the directed technical change and induced innovation literatures (Popp, 2002; Acemoglu et al., 2012; Calel and Dechezleprêtre, 2016; Calel, 2020; Casey, 2024), providing firm-level evidence that carbon pricing itself redirects investment toward cleaner production technologies.[10]

and find leakage rates between -0.7–2.2% across five imputation strategies in the five-year counterfactual with clean technology adoption.

[9] Appendix C shows that this representation holds much more generally under standard regularity conditions, following the older literature going back to Pethig (1976); Siebert et al. (1980); Baumol and Oates (1988) and Cropper and Oates (1992).

[10] Unlike our findings across multiple countries from 2005 to 2021, Calel (2020) finds limited adoption of clean

Although we focus on EU-ETS in this paper, the framework is portable to other settings: wherever firm-level emissions and financial data are available, researchers can evaluate the aggregate and distributional effects of pollution pricing across a range of regulatory contexts.

2 Institutional Setting and Data

2.1 EU Emissions Trading System

The European Union Emissions Trading System (EU-ETS), established in 2005, is the world's first international carbon market and the cornerstone of EU climate policy. EU-ETS is a cap-and-trade system that sets a total limit on greenhouse gas emissions for regulated sectors and reduces this cap over time. Within this framework, firms receive or purchase emission allowances that permit emission of one metric ton of CO_2. The allowances can be traded, which provides incentives for companies to reduce their emissions. Firms that cut emissions can sell surplus allowances and raise revenue, while those exceeding their limits must buy additional allowances and incur costs. Compliance is strictly monitored and regulated installations must report their greenhouse gas emissions annually. These annual reports are subject to independent third-party verification.

EU-ETS has undergone multiple phases and reforms. Phase I (2005–2007) was a pilot phase establishing baselines and market infrastructure. Phase II (2008–2012) coincided with the global financial crisis, and had depressed levels of emissions and a large surplus of allowances, depressing permit prices. In response, the EU implemented backloading and the Market Stability Reserve (MSR). Backloading delayed the auction of 900 million allowances, and the MSR, which began operation in 2019, adjusts auctioned allowance supply based on the total number of unused allowances in circulation to stabilize allowance prices. A 2018 reform strengthened the MSR, contributing to a substantial increase in permit prices that persisted through the end of our sample period. These policy reforms provide one source of exogenous supply-driven variation in permit prices that we exploit for identification of firm production function parameters.

In its early years, most allowances were allocated for free, but the system has gradually shifted toward auctioning. The scope of EU-ETS has also expanded over time: while the program initially covered power generation and energy-intensive manufacturing, Phase III (2013) extended it to additional industrial sectors, notably aluminum, petrochemicals, and ammonia and nitric-acid production.

2.2 Data Sources

Here we describe our data sources. Appendix B provides more details on how we use these data to construct variables for estimation and simulation.

technology in UK firms using data through 2012. Given EU-ETS began in 2005 and our finding that clean technology takes years to adopt, these results are not inconsistent with one another.

Firm-Level Financials We use data from Bureau van Dijk's Orbis dataset to measure firm-level economic outcomes. Orbis aggregates information from administrative records and private surveys to construct a harmonized dataset for both private and public companies. The dataset includes balance sheet, income statement, and employment data. Variables relevant for our estimation include the cost of labor, fixed assets, employment, and operating revenues. For each firm, Orbis also reports its NACE sector code, country, and address. Orbis has substantial coverage in Europe, and, in some countries such as Spain, has close to universal coverage of firms.

Materials Quantities and Prices Orbis reports materials spending but not physical quantities. We recover firm-level materials quantities by dividing reported firm-level materials costs by the country-sector-year intermediate input price index P_{nt}^{M} from EU KLEMS. These quantities enter our estimation of the elasticity of substitution among capital, labor, and materials, as well as the simulation. We also use these price indices as an instrument for the cross-sector demand elasticity.

Input-Output Structure and Final Demand We use Eurostat Supply-Use tables to measure two objects that enter the counterfactuals. First, intermediate use flows across sectors are used to calibrate the sector coefficients in a producer's materials bundle. Second, final demand expenditures are used to calibrate the final demand shares across sectors for the demand system. The same final demand expenditures also serve as the outcome variable when we estimate the cross-sector demand elasticity, with the country-sector-year gross output price index from EU KLEMS as the price facing households.

Capital Rental Prices and Deflators We construct capital rental prices using a Jorgenson formula and data from EU KLEMS capital accounts and Penn World Table (PWT) 11.0 (Feenstra et al., 2015) country-year real returns and price indexes. The same PWT price information is used to express revenues, input expenditures, fixed assets, value added, the intermediate input price index, and emissions prices in real terms.[11]

Emissions We obtain installation-level emissions from the European Union Transaction Log (EUTL). The EUTL records and tracks permit transactions and verified emissions covered by EU-ETS. Each installation is a physical entity regulated by EU-ETS. For each installation-year, the EUTL reports its name and location, verified emissions, and permits allocated, traded, and surrendered.

Crosswalk We link the Orbis financial data to the EUTL emissions data using the EUTL-Orbis crosswalk provided by the European Commission's Joint Research Centre Data Catalogue (Letout, 2021).

[11] Because the deflator is country-year specific, the real permit price can vary across countries even though the nominal allowance price is common to all firms. This variation is, however, negligible.

Carbon Prices We obtain annual average EU-ETS permit prices from the World Carbon Pricing Database maintained by Resources for the Future (RFF). This database compiles carbon pricing data for both cap-and-trade programs and carbon taxes globally. Since the Orbis data are annual, we use the average transaction price for EU-ETS permits as our measure of the permit price for each year.

Carbon Price Surprise Series We obtain the high-frequency carbon price surprise series of Känzig (2025), extended by Jézéquel-Royer and Levieuge (2025). The series measures unexpected changes in EU-ETS permit prices around discrete regulatory announcements—European Commission auction decisions, Parliament votes, and court rulings—by computing the change in carbon futures prices in a narrow window around each event. We aggregate the daily series to annual frequency and use it as an instrument for the permit price in our estimation.

2.3 Sample Construction

We match 57% of verified EU-ETS emissions to Orbis firms, then apply common accounting data screens—chiefly requiring positive reported materials costs, which several countries' Orbis filings omit entirely. In our primary matched EUTL-Orbis panel we have 4,671 regulated firms across 24 countries from 2005 to 2021, for which we observe both firm-level financials and verified emissions, still covering 29% of emissions under EU-ETS. This matched sample underlies our main estimation and counterfactuals. The one exception is the capital-labor-materials elasticity of substitution: because it is identified from wage variation and does not require verified emissions, we estimate it on a broader Orbis panel spanning 1998–2021 via a granular instrumental variable (GIV) design. These regressions contain 25.8 million observations in total; Appendix F describes the construction of the instrument.

We also construct a larger Orbis panel that retains all Orbis firms in our sectors regardless of whether they are linked to an ETS installation, starting from the same countries and years as our primary matched dataset. This panel contains 24 million firm-year observations. We use it in an extended version of the model to quantify how EU-ETS leads to emissions leakage to unregulated firms through reallocation of production. Because verified emissions are observed only for firms linked to EU-ETS installations, the leakage extension requires us to impute emissions for non-ETS firms. Appendix D.6 provides the full imputation procedure.

Power dominates the matched sample in terms of emissions. Together with cement and glass, refining, basic metals, and chemical products, these five largest emitting sectors account for approximately 86% of matched-sample emissions, with the remainder spread across a range of manufacturing and non-manufacturing activities. Appendix Table A1 reports the full sectoral breakdown.

Because the EU-ETS regulates *installations* based on their physical activity—combustion above 20 MW, cement production, metal ore sintering, and so on—rather than firms based on their industry classification, a firm's Orbis NACE code may not reflect the activity of its regulated installation. For example, a construction company may operate an asphalt mixing plant, a holding company may run a power plant, and an oil extraction firm could own a refinery. We classify firms by their primary Orbis NACE code. This captures the firm's primary economic activity, which better represents the economic decision-making (and financial data) for the firm than the physical activity of an individual regulated installation. This yields 11 sectors: six correspond to directly regulated EU-ETS activities (power, cement and glass, basic metals, refining, chemical products, and pulp and paper), two retain distinct manufacturing identities (rubber and plastics, fabricated metals), mining and quarrying forms its own group, and the remaining sectors are consolidated into light manufacturing and other non-manufacturing.

To get a sense of whether this classification is economically appropriate, we compare firms' Orbis sectors to the sector of the matched installations in the EUTL. For the five largest emitting sectors, a firm's 2-digit Orbis-reported NACE sector and the 2-digit NACE sector implied by its EUTL installations agree 87.2% of the time in chemicals at the low end, up to 99.7% of the time at the high end for power when weighting by emissions. Agreement is slightly lower in the five sectors that do not correspond to a directly regulated activity (light manufacturing, mining and quarrying, rubber and plastics, fabricated metals, and other non-manufacturing). For these sectors, the Orbis sector is the economically relevant one for about three-quarters of firms. These firms operate in a genuine production sector that matches their Orbis sector, while their regulated installation is typically an on-site combustion or power unit. Classifying these firms by their installation sector would misclassify manufacturing or mining firms as power producers. The remaining cases are concentrated in our residual other-non-manufacturing category. These are typically firms whose Orbis code reflects a legal parent, holding company, or trading entity.

Appendix Figure A1 shows how sectoral emissions shares have evolved over time. While power generation has consistently dominated emissions, its share has declined from 2005 to 2021, reflecting improvements in power sector efficiency and the relative growth of other regulated sectors.

3 Model

In our model i indexes firms, n indexes sectors, and t indexes years. We use $\mathcal{N}$ to denote the set of sectors. Our model has a short run and a long run. In the short run, monopolistically competitive firms take clean technology as given and choose inputs and output prices to maximize profits. The short run captures firms' input substitution margin and reallocation of output across firms with heterogeneous emissions intensities. In the long run, firms invest in clean technology, which accumulates through a reduced-form law of motion and alters the production technology that enters future short-run equilibria.

3.1 Production and Abatement

Firms produce output using capital, labor, and materials. Production generates emissions as a byproduct that can be reduced through costly abatement. The challenge for estimation is that abatement effort is typically unobserved. Researchers often observe firms' emissions and their use of capital, labor, and materials, but not how they allocate resources toward abatement activities. Our approach addresses unobserved abatement by showing that, under a weak set of assumptions on production, the underlying model is equivalent to a production function where emissions enter as an input. This equivalence means we can apply standard production function estimation techniques directly on observed inputs and emissions without needing to observe abatement. In our setting we impose a nested constant elasticity of substitution functional form.

Short-run production proceeds in two stages. First, firms combine capital, labor, and materials into a productive composite. Second, they choose how much of this productive capacity to allocate toward output versus abatement. In the first stage, firm i in sector n at time t produces the composite:

$$X_{int} = \left[\delta_{K,in}\left(K_{int}\right)^{\frac{\xi_n-1}{\xi_n}} + \delta_{L,in}\left(L_{int}\right)^{\frac{\xi_n-1}{\xi_n}} + \delta_{M,in}\left(M_{int}\right)^{\frac{\xi_n-1}{\xi_n}}\right]^{\frac{\xi_n}{\xi_n-1}}. \tag{1}$$

This composite represents a firm's productive capacity, the total resources it can devote to output and abatement. The δ parameters are share parameters that sum to one across inputs, and ξ_n is the elasticity of substitution between capital K_{int}, labor L_{int}, and materials M_{int}.

Firms abate by diverting a share of their productive capacity toward abatement activities rather than output. Let $\vartheta_{int} \in [0, 1)$ denote the fraction of the productive composite allocated to abatement rather than output. For example, a steel plant may substitute scrap metal for virgin ore to avoid the ore reduction that generates most of its direct emissions, but at the cost of additional capital, labor, and materials inputs for processing. Conditional on the input and abatement choice, net output is:

$$Y_{int} = (1 - \vartheta_{int})A_{int}X_{int} \tag{2}$$

where A_{int} is Hicks-neutral productivity. This formulation is directly analogous to the "share of output devoted to abatement" approach in the literature (e.g. Copeland and Taylor, 2004; Shapiro and Walker, 2018; Hollingsworth et al., 2026).

The remaining question is how abatement effort translates into emissions reductions. We model this through an abatement function $\psi(\vartheta_{int})$ that maps abatement intensity to emissions per unit of the productive composite:

$$\frac{E_{int}}{X_{int}} = \psi(\vartheta_{int}; \pi_{in}, \sigma_n, B_{int}).$$

The function ψ is decreasing and convex in ϑ_{int} and B_{int}. We adopt a functional form governed by three parameters:

$$\psi(\vartheta_{int}) = \frac{1}{B_{int}}\left(\frac{(1-\vartheta_{int})^{\frac{\sigma_n-1}{\sigma_n}} - (1-\pi_{in})}{\pi_{in}}\right)^{\frac{\sigma_n}{\sigma_n-1}}. \tag{3}$$

Here B_{int} is the firm's *clean technology*, σ_n governs the curvature of abatement costs, and π_{in} is a share parameter.

Combining equations (2)–(3) yields an equivalent representation of production where emissions appear as an input to producing net output rather than a byproduct:

$$Y_{int} = A_{int}\left[(1-\pi_{in})X_{int}^{\frac{\sigma_n-1}{\sigma_n}} + \pi_{in}\left(B_{int}E_{int}\right)^{\frac{\sigma_n-1}{\sigma_n}}\right]^{\frac{\sigma_n}{\sigma_n-1}}. \tag{4}$$

In this representation, the abatement function parameters take on a production function interpretation. The parameter σ_n becomes the *elasticity of substitution* between the productive composite and emissions—it determines how easily firms can reduce emissions by reallocating inputs towards abatement efforts. When σ_n is low, emissions and productive inputs are near-complements and firms can only make deep emissions cuts by substantially reducing output.[12] Appendix Figure A2 illustrates this relative price logic, with Panel A2a showing the initial input choice and Panel A2b showing the response to an emissions price increase.

The share parameter π_{in} determines the baseline emissions intensity of production. In the Cobb-Douglas special case, π_{in} determines the cost share of emissions. B_{int} is an emissions-saving technology parameter that we call *clean technology.* It multiplies emissions in equation (4), so higher B_{int} means a firm can produce the same level of output with fewer emissions. For example, a firm with gas-fired power plants has higher B_{int} than a firm with coal plants but identical balance sheet information because it emits roughly half the CO_2 per kWh given the same observed levels of capital, labor, and materials.[13]

ϑ_{int} is a firm's unobserved within-period choice of abatement intensity conditional on its existing state of technology, while B_{int} reflects predetermined differences in technology. In the long run, firms can invest in clean technology and raise B_{int}, effectively relaxing the trade-off between emissions and the productive composite use in yielding net output (Section 3.4). This division gives the model a "putty-clay" like structure (Johansen, 1959; Atkeson and Kehoe, 1999): σ_n is the *ex post* elasticity, governing the substitution/abatement possibilities available to a firm with equipment it has in place, while investment in B_{int} is the *ex ante* margin that selects among all the technology and abatement possibilities available to the firm.[14]

The reformulation of emissions as an input is the key to estimation: equation (4) contains quantities we observe or can recover—the productive composite and emissions—even though the underlying model involves unobserved abatement effort ϑ_{int}. In Section 4, we show how to use the firm's conditional factor demands to estimate the nested CES production model using only factor prices and input quantities.

Although we adopt a nested CES specification, the emissions-as-input equivalence holds more generally: Appendix C proves it under standard regularity conditions—relaxing constant elas-

[12] For example, cement production releases CO_2 directly through calcination—a chemical reaction, not combustion—so no reallocation of inputs can avoid these emissions, implying a low σ_n. Electricity generation may have higher σ_n at plants that have the ability to combust and substitute amongst multiple types of fuels such as coal and gas or gas and fuel oil.

[13] Similar to a Solow residual, clean technology captures how firms with the same input expenditures on the balance sheet have different emissions intensities because of unobserved heterogeneity in the composition and emissions of these inputs.

[14] One important distinction that makes the connection to a "putty-clay" model not exact is that B_{int} reflects a productivity term as opposed to a capital vintage as is the case in most of these types of models.

ticities, constant returns to scale, and the separability of abatement and production—and Appendix A.1 tests the nested CES restriction with functional-form-free Morishima elasticities. We adopt the nested CES for estimation and counterfactuals because it fits the data well.

3.2 Short Run Equilibrium

We begin with the short run equilibrium of our model. Our key timing assumption is that firms' productivities A_{int} and B_{int} are predetermined and fixed at the time input choices are made. Firms can endogenously change clean technology B_{int} in response to carbon prices, but these adjustments take time and occur across periods rather than within one. We model this long-run response in Section 3.4. Within each period, firms minimize costs over inputs taking factor prices as given, set prices as constant markups over marginal cost given monopolistic competition and CES demand, and goods markets clear.

3.2.1 Firm Cost Minimization

Firm i in sector n is a price-taker in input markets and faces prices P^K_{int}, P^L_{int}, P^M_{int}, and P^E_t for capital, labor, materials, and emissions. Taking factor prices and technologies (A_{int}, B_{int}) as given, it chooses inputs to minimize variable costs conditional on output Y_{int}:[15]

$$\min_{X_{int},E_{int}} \left\{ P^X_{int} X_{int} + P^E_t E_{int} \;\middle|\; Y_{int} = A_{int} \left[(1-\pi_{in}) X_{int}^{\frac{\sigma_n - 1}{\sigma_n}} + \pi_{in} \left(B_{int} E_{int} \right)^{\frac{\sigma_n - 1}{\sigma_n}} \right]^{\frac{\sigma_n}{\sigma_n - 1}} \right\}$$

where P^X_{int} is the CES unit cost of the composite input X_{int}. The first-order conditions imply a log-linear relationship between input ratios and relative prices. Expressing this in terms of the emissions-to-composite ratio—the natural measure of emissions intensity in our model—yields:

$$\underbrace{\log \frac{E_{int}}{X_{int}}}_{\text{Emissions Intensity}} = \underbrace{\sigma_n \log \left(\frac{\pi_{in}}{1-\pi_{in}} \right)}_{\text{Constant}} - \sigma_n \underbrace{\log \frac{P^E_t}{P^X_{int}}}_{\substack{\text{Emissions}\\ \text{Composite}\\ \text{Price Ratio}}} + (\sigma_n - 1) \underbrace{\log B_{int}}_{\substack{\text{Clean}\\ \text{Tech}}}. \tag{5}$$

The coefficient $-\sigma_n$ is the strength of the short-run input substitution margin: it governs how firms' emissions intensity responds to a change in the relative price of emissions, holding clean technology B_{int} fixed. Higher emissions prices lower E/X, but when σ_n is small the response is limited. Clean technology also lowers E/X when emissions and the productive composite are complements ($\sigma_n < 1$).

The analogous inner nest conditions govern substitution among capital, labor, and materials,

[15] We omit the analogous inner nest problem of minimizing unit cost for X_{int}.

one for each input pair. Here we display the materials-to-labor ratio:

$$\underbrace{\log \frac{M_{int}}{L_{int}}}_{\text{Input Ratio}} \quad = \quad \underbrace{\xi_n \log\left(\frac{\delta_{M,in}}{\delta_{L,in}}\right)}_{\text{Constant}} \quad - \quad \xi_n \underbrace{\log \frac{P^M_{int}}{P^L_{int}}}_{\substack{\text{Materials}\\ \text{Labor}\\ \text{Price Ratio}}} \,. \tag{6}$$

Higher labor prices induce firms to substitute toward materials, raising M/L; the magnitude of this response identifies ξ_n.

3.2.2 Demand and Markups

Product demand has two layers. Within each sector, a competitive sectoral composite producer bundles firm varieties into a sectoral composite using CES technology with within-sector demand elasticity ϵ_n. Households then allocate exogenously fixed final expenditure across sectoral composites through a second CES nest with cross-sector demand elasticity υ and a corresponding final demand price index P^F_t. We specify the full demand system in Appendix D. Because firms also purchase sectoral composites as materials inputs (Section 3.2.3), total sector demand includes both household final demand and materials demand from downstream firms.

Facing CES demand, monopolistically competitive firms optimally set prices as a constant markup μ_n over marginal cost:

$$\mu_n = \frac{P_{int}}{MC_{int}} = \frac{\epsilon_n}{\epsilon_n - 1}. \tag{7}$$

The firm's problem can thus be decomposed into minimizing costs conditional on output, and setting price as a constant markup over marginal cost and letting the demand system determine its output level.

3.2.3 Input-Output Linkages

Input-output linkages are central in this setting because ETS-regulated sectors produce substantial amounts of materials for use by firms, and they are also users of carbon-intensive materials from other ETS-regulated sectors. For example, electricity and refined petroleum products from carbon-intensive power generators and refineries are inputs into manufacturing and cement. A carbon price that raises the costs of upstream energy sectors is passed-through to downstream manufacturers' marginal costs through their materials input costs. This amplifies and propagates the negative economic effects of the carbon price. It also induces reallocation, both within and across sectors, beyond the direct effects of carbon pricing.

We model firms' materials input in equation (1) as a Cobb-Douglas aggregate of sectoral composites:

$$M_{int} = \varrho_{in}^{-1} \prod_{m \in \mathcal{N}} \left(\frac{M_{imnt}}{\Omega_{mnt}}\right)^{\Omega_{mnt}}. \tag{8}$$

The aggregator is sector-specific with materials expenditure shares Ω_{mnt} for user sector n buying materials from supplier sector m, where $\sum_{m\in\mathcal{N}}\Omega_{mnt} = 1$. The aggregator has a firm-specific shifter $\varrho_{in} > 0$ allowing baseline materials prices to differ across firms within a sector. The implied firm-level materials unit costs are:

$$P^M_{int} = \varrho_{in} P^M_{nt} \qquad \text{where} \qquad P^M_{nt} = \prod_{m\in\mathcal{N}} (P_{mt})^{\Omega_{mnt}}. \tag{9}$$

3.2.4 Goods Market Clearing

Let $S_{nt} \equiv P_{nt}Y_{nt}$ denote total sector sales and let $S^F_{nt} \equiv P_{nt}Q^F_{nt}$ denote household final expenditure on sector n. Goods market clearing requires that gross sector sales equal final demand plus materials demand from downstream sectors:

$$S_{nt} = S^F_{nt} + \sum_{m\in\mathcal{N}} \Omega_{nmt}\alpha^M_{mt}S_{mt}, \tag{10}$$

where $\alpha^M_{nt} = \mu_n^{-1}\sum_i \frac{P_{int}Y_{int}}{S_{nt}} s_{M,int}$ is materials expenditure as a share of sector n sales and $s_{M,int}$ is the share of materials in firm i's variable costs (see Appendix D).

In our main model we do not model the labor or capital markets and instead assume there is a perfectly elastic supply of both inputs. Appendix E introduces a full general equilibrium version of the model with markets for labor, capital, and emissions, and a fixed aggregate amount of each.

3.2.5 Equilibrium

An equilibrium in each period t consists of firm-level prices, quantities, and input choices $\{P_{int}, Y_{int}, X_{int}, E_{int}, K_{int}, L_{int}, M_{int}, \{M_{imnt}\}_{m\in\mathcal{N}}\}_{i,n}$, sectoral objects $\{P_{nt}, Y_{nt}, S_{nt}, P^M_{nt}\}_n$, household final demand quantities $\{Q^F_{nt}\}_n$, and the final demand price index P^F_t such that: (i) each firm minimizes costs given factor prices and technologies (A_{int}, B_{int}); (ii) firms set prices as constant markups over marginal cost; (iii) sectoral composite producers and households satisfy the CES demand systems in Section 3.2.2, given exogenous aggregate household expenditure; (iv) materials prices satisfy equation (9); and (v) goods markets clear according to equation (10).

Capital and labor prices are fixed and firms face a common emissions permit price P^E_t; materials prices and gross sectoral sales are endogenous via the input-output fixed point. This equilibrium representation preserves supply-chain pass-through while abstracting from movements in primary factor prices.

3.3 Aggregate Elasticities and Marginal Abatement Costs

Equation (5) showed that σ_n governs how a single firm's emissions intensity responds to the price of emissions. Aggregate emissions also fall in the short run for a second reason: carbon pricing raises unit costs more at firms that are emissions-intensive, or use materials from dirty suppliers, compared to firms that emit little or source clean inputs. Through the demand system of Section 3.2.2, this change in relative unit costs across firms moves output and market share toward cleaner producers. Quantifying what a carbon price achieves therefore requires an aggregate sector-level analogue to σ_n that captures both.

This subsection shows how to construct the aggregate sector-level elasticity of substitution between emissions and the productive composite $\bar{\sigma}_{nt}$ following Oberfield and Raval (2021). Unlike σ_n, it is not a technology parameter but a statistic of the sector's composition, which is why it carries a time subscript. We then show how the elasticity determines aggregate marginal abatement cost curves.

A sector's ease of abatement has two components: substitution away from emissions within the firm, and reallocation of output across firms. When permit-cost pass-through—how strongly the permit price raises a firm's composite input price through its supply chain—is common across firms or uncorrelated with their emissions cost shares, Appendix H.2 shows that the aggregate elasticity is a simple convex combination of the two components:[16]

$$\bar{\sigma}_{nt} = \underbrace{(1-\chi_{nt})\sigma_n}_{\text{within-firm}} + \underbrace{\chi_{nt}\epsilon_n}_{\text{reallocation}} . \tag{11}$$

A high $\bar{\sigma}_{nt}$ means the sector cuts emissions readily when the carbon price rises while a low one means the sector must shrink for emissions to fall.

The reallocation component is the product of the within-sector demand elasticity ϵ_n and the emissions heterogeneity index $\chi_{nt} \in [0,1]$, which measures the dispersion in emissions intensity across firms. We make χ_{nt} precise below. Both factors are necessary for reallocation. Dispersion is what makes a common carbon price a *relative* price change, raising unit costs by more at dirty firms than at clean ones.[17] Elastic demand is what turns that cost gap into a shift in market shares. If firms are alike, a carbon price leaves relative prices unchanged, so no shares move, regardless of how elastic demand may be. If demand is inelastic, relative prices move but buyers barely respond so production does not shift. Only when we have dispersion in emissions intensity and elastic demand does output move toward cleaner producers. In this case, sector emissions fall even without any firm changing its technology or its input mix.

The within-firm component is governed by σ_n, carrying the complementary weight $1-\chi_{nt}$. If every firm in the sector is equally dirty, $\chi_{nt}=0$ and $\bar{\sigma}_{nt}=\sigma_n$: every ton abated has to come from within firms. As firms grow more heterogeneous, χ_{nt} rises and opportunities for reallocation grow. σ_n sets what a firm can do on its own while χ_{nt} determines how much that ability matters for the sector versus reallocation, which is governed by ϵ_n. If firms differ and demand is sufficiently elastic ($\epsilon_n > \sigma_n$), reallocation can make the sector more elastic than any individual firm: $\bar{\sigma}_{nt} > \sigma_n$.

We now formally define χ_{nt}. Let $s_{E,int} \equiv P_t^E E_{int}/(P_t^E E_{int} + P_{int}^X X_{int})$ be firm i's emissions cost

[16] When pass-through instead covaries with firms' emissions cost shares, the elasticity acquires a covariance correction term. Our results use this corrected elasticity (Appendix H.2). The correction is modest: across sectors, the median absolute change in the aggregate elasticity is 0.01 (9%), and the largest is 0.09 (19%) in refining.

[17] From Shephard's lemma, cost shares—or emissions intensities with common permit prices in our case—tell us how costs respond to permit prices. These cost increases then get translated into consumer-facing output prices with a constant markup.

share. χ_{nt} is the cost-weighted cross-sectional variance of these shares, normalized to lie in $[0,1]$:

$$\chi_{nt} \;\equiv\; \frac{\sum_i w_{i|n,t}\,(s_{E,int}-\Gamma_{nt})^2}{\Gamma_{nt}\,(1-\Gamma_{nt})} \;\in\; [0,1], \tag{12}$$

where $\Gamma_{nt} = \sum_i w_{i|n,t}\, s_{E,int}$ is the sector emissions cost share and $w_{i|n,t}$ is firm i's share of total sector variable cost. The numerator is the dispersion in how emissions-intensive firms are, and the denominator is the most dispersion a sector with average intensity Γ_{nt} could display, so the index is zero when every firm has the same emissions cost share and approaches one when those shares are maximally spread.

The index also has a direct reading in terms of clean technology. Under Leontief production ($\sigma_n \to 0$)—which we find empirically (Section 5.1.2)—a first-order approximation reduces it to:[18]

$$\chi_{nt} \;\approx\; \underbrace{\Gamma_{nt}\,(1-\Gamma_{nt})}_{\substack{\text{Sector-level}\\ \text{cost share normalization}}} \quad \underbrace{\mathrm{Var}_i^s(\log B_{int})}_{\substack{\text{Cross-firm dispersion}\\ \text{in clean technology}}} \;,$$

where Var_i^s denotes the variance across firms, taken with the same cost weights as in equation (12). To a first-order, clean technology dispersion is a primary factor in driving reallocation.

The same construction applies one level up for reallocation *across* sectors. The economy-wide aggregate elasticity combines abatement within sectors with reallocation of spending across them, weighted by a cross-sector heterogeneity index $\chi_t \in [0,1]$ that measures dispersion in emissions intensity across sectors:

$$\bar{\sigma}_t = \underbrace{(1-\chi_t)\bar{\sigma}_t^N}_{\text{within-sector}} \quad + \quad \underbrace{\chi_t \upsilon}_{\substack{\text{cross-sector}\\ \text{reallocation}}} \;, \tag{13}$$

where $\bar{\sigma}_t^N$ aggregates the sectoral substitution elasticities through the input-output network and υ is the cross-sector demand elasticity. Emissions and costs here are embodied: each supply chain's emissions are attributed to the final good containing them, so a sector's carbon exposure includes what its suppliers pass down.[19]

From elasticities to marginal abatement costs These elasticities also describe the marginal abatement cost curves commonly used in environmental policy analysis. The firm, sector, or economy-wide marginal abatement cost (MAC) is the shadow cost of cutting emissions while holding output and clean technology fixed at the corresponding level of aggregation. At the economy-wide level the fixed aggregate is household final demand (Appendix H.3). We show in Appendix H that the elasticity of a MAC with respect to emissions is inversely proportional to the elasticity of

[18] Appendix H.2.5 derives the approximation. The approximation error is governed by the dispersion of $\log B_{int}$, not by σ_n, and by omitted cross-firm variation in composite prices P_{int}^X, production function weights π_{in}, and base-year emissions E_{in0}.

[19] Appendix H.3 derives the exact economy-wide response used in our results: the within-good weight gains a covariance between embodied emissions shares and pass-through, and a markup covariance enters the response. At the observed allocation both corrections are negligible, moving the response by 0.1% relative to equation (13).

substitution. In cases where σ_n is small, firm MAC schedules are close to vertical. On the other hand, higher reallocation potential (via elastic demand and heterogeneous firms) increases $\bar{\sigma}_{nt}$ and flattens the sector MAC. Steep firm MAC schedules can co-exist with flat aggregate ones.[20]

MAC curves are often built from the bottom up, by stacking plant- or technology-level abatement costs, or by treating a sector as a representative firm. This omits reallocation and the cost propagation that arises through the input-output network. Our MAC construction generalizes these standard approaches by incorporating reallocation and supply chain impacts. The underlying elasticities are also estimated from firm-level data rather than assumed from technology costs.

3.4 Long Run Clean Technology Dynamics

Section 3.2 explained how firms choose inputs given predetermined clean technology B_{int}. Over longer horizons, firms invest to change B_{int}, altering future short run equilibria. Rather than fully specifying the firm's dynamic investment problem, we adopt a flexible semiparametric approach that traces out how clean technology evolves in response to an exogenous change in the carbon price. A long run equilibrium is the short-run model of Section 3.2 with B_{int} replaced by this estimated path.[21]

Under a time-to-build assumption, investment $I^B_{in,t-1}$ chosen at date $t-1$ raises clean technology in period t:

$$\log B_{int} = \log B_{in,t-1} + I^B_{in,t-1}, \tag{14}$$

so the cumulative change in $\log B$ from $t-1$ through $t+h$ is the sum of interim investments:

$$\log B_{in,t+h} - \log B_{in,t-1} = \sum_{r=0}^{h} I^B_{in,t-1+r}. \tag{15}$$

We model firm investment with a linear reduced-form rule in which the date $t-1$ carbon price and composite input price shift clean technology investment around a firm-specific steady state rate $\bar{I}^B_{in}$:

$$I^B_{in,t-1} = \bar{I}^B_{in} + \beta^E_n \log P^E_{t-1} + \beta^X_n \log P^X_{in,t-1} + \eta_{in,t-1}, \tag{16}$$

where $\eta_{in,t-1}$ is a mean-zero residual and β^E_n, β^X_n are sector-specific reduced form coefficients. This approach is an alternative to specifying the firm's full dynamic investment problem. Substituting equation (16) into equation (15) and projecting on the date $t-1$ prices yields the reduced form local projection for clean technology, where the two-regressor form treats expectations of future log

[20] Generally, this is true only if demand at each level is more elastic than the substitution margin it aggregates, which aligns with our estimates.

[21] Appendix I derives the linear rule as a first-order approximation to the policy function of a general dynamic investment problem, under linear price forecasts, and shows that the theory of directed technical change signs its coefficient given our estimated σ_n. A one-period implementation lag and no depreciation of the clean technology stock deliver both the log-additive law of motion and the local projection specification below.

prices as linear in the date-$t-1$ log prices and the investment residuals as orthogonal to them:

$$\mathbb{E}_{t-1}[\log B_{in,t+h} - \log B_{in,t-1}] = \alpha_{in}^{(h)} + \Psi_{h,n}^{E} \log P_{t-1}^{E} + \Psi_{h,n}^{X} \log P_{in,t-1}^{X}. \tag{17}$$

The horizon-h coefficient $\Psi_{h,n}^{E}$ is the cumulative impulse response of clean technology to a permit price shock, combining the direct effect of $\log P_{t-1}^{E}$ on $I_{in,t-1}^{B}$ with the indirect effect through expectations over future prices and subsequent investment. Section 4.3 describes how we estimate the reduced-form coefficients $\{\Psi_{h,n}^{E}\}$ using an observable empirical proxy for B_{int}.[22]

The estimated path of clean technology also shifts the MAC schedules of Section 3.3. Because a higher B_{int} lets a firm produce the same output with fewer emissions, it lowers that firm's emissions cost share $s_{E,int}$ at any given permit price; we derive the closed-form firm MAC in Appendix H.1 and show there that cleaner technology shifts the fixed-output schedule downward. Aggregating across firms and sectors, clean technology investment therefore shifts the level of the economy-wide MAC schedule downward while leaving its elasticity, which equation (13) governs, approximately unchanged, as Panel (b) of Figure 5 in Section 5.4 shows.

3.5 Extension: Leakage to Unregulated Firms

Our main model focuses on EU-ETS-regulated firms since emissions at non-ETS firms are unobserved. Since unregulated firms compete with regulated ones in the same goods markets, carbon pricing may shift production toward unregulated producers and undo some of the intended emissions reductions. This phenomenon is called carbon leakage. We quantify an extended version of the model using the full Orbis set of firms to measure the extent of carbon leakage. Unregulated firms face the same demand system and input-output network as those under EU-ETS, but do not pay the permit price on their emissions, so their privately optimal abatement choice is zero and output reduces to $Y_{int} = A_{int} X_{int}$. They are still affected by policy indirectly: materials costs rise through upstream pass-through, while demand shifts toward them as regulated competitors become more expensive. These indirect effects change production at non-ETS firms in our counterfactual simulations, which we interpret as leakage. Since emissions at these firms are unobserved, we impute emissions intensities using several different strategies.

4 Estimation

We organize the estimation around the three margins developed in Section 3. Section 4.1 estimates the production elasticities ξ_n and σ_n that identify the short-run input substitution margin. Section 4.2 recovers the emissions heterogeneity index χ_{nt} and the demand elasticities ϵ_n and υ. χ_{nt} and ϵ_n pin down the reallocation margin and the marginal abatement cost curves at the sector-

[22] Appendix K derives the long-run elasticity of substitution between emissions and the productive composite. When emissions and the composite are Leontief like we estimate below, the long-run elasticity of substitution is just the elasticity of clean technology with respect to the permit price $\Psi_{h,n}^{E}$ adjusted for the extent of pass-through of permit prices through the input-output network.

level, while v comes in at the economy-wide level. Section 4.3 uses local projections to estimate the cumulative clean technology impulse response $\Psi^E_{h,n}$. It tells us how B_{int} responds to permit price shocks over time, which governs the long run adjustment margin. Throughout, our identifying variation in permit prices comes from the carbon price surprise series.

4.1 Production Elasticities

We estimate the two production elasticities ξ_n and σ_n from firms' conditional factor demands. σ_n measures how easily firms substitute away from emissions at a given B_{int}, while ξ_n measures substitutability among capital, labor, and materials.

4.1.1 Inner Nest: Capital-Labor-Materials (ξ_n)

Under CES production, the optimal ratio of any two inputs j and k satisfies a relationship analogous to equation (6), governed by the common elasticity ξ_n. Our estimating equation for the materials-to-labor ratio is:

$$\log \frac{M_{int}}{L_{int}} = \xi_n \log P^L_{r,int} + \beta_M \log P^M_{int} + \lambda_{1,in} + \lambda_{1,ct} + \varepsilon_{1,int}, \tag{18}$$

where $\lambda_{1,in}$ is a firm fixed effect, $\lambda_{1,ct}$ is a country-year fixed effect that absorbs national wage and price trends, and $\varepsilon_{1,int}$ is the error term. The wage $P^L_{r,int}$ is our variable of interest and is endogenous. We label it with an additional subscript r because we will be using a regional wage consistent with our regional labor market instrumental variables approach.

The ξ_n coefficient on the log wage is the elasticity of substitution. When wages rise, firms substitute toward materials, and the extent of this substitution identifies ξ_n. The identifying variation we use is within-country-year variation in regional wages. We estimate the regression separately for each sector n and we two-way cluster standard errors by firm and year.[23] The CES structure implies that we can also estimate this elasticity from the capital-to-labor ratio, $\log(K/L)$, which yields the same ξ_n if the nested CES specification is correct. We estimate both and report the results in Appendix A.

The identification challenge is that wages are endogenous. Local wages may comove with unobserved shocks, such as productivity shocks, that also affect firms' input choices. We address this using a granular instrumental variable (GIV) strategy following Gabaix and Koijen (2024). The intuition is that since the firm size distribution in European labor markets is right-skewed, idiosyncratic labor demand shocks to large employers bid up local wages for all firms in the region, acting as residual labor supply shifters uncorrelated with any individual firm's productivity. We define labor markets at the region-year level and construct a leave-one-out instrument that aggregates idiosyncratic employment shocks to other firms, weighted by their employment shares. Technical details of the GIV construction are provided in Appendix F.

[23] Under CES, the coefficient on $\log P^M$ could also identify ξ_n. We focus on $\log P^L$ because our granular instrumental variable strategy provides a credible instrument for wages but not for materials prices.

4.1.2 Outer Nest: Emissions-Composite (σ_n)

We estimate the outer nest elasticity σ_n between emissions and the productive composite using the analogous conditional factor demands in equation (5). Given the inner nest estimates ξ_n, we construct the composite price P^X_{int} as the normalized nested-CES price index over capital, labor, and materials. CES weights are given by baseline cost shares and substitution is governed by ξ_n. We measure the productive composite quantity as observed non-emissions spending deflated by the price index: $X_{int} = (C^K_{int} + C^L_{int} + C^M_{int})/P^X_{int}$.

Our estimating equation is:

$$\log \frac{E_{int}}{X_{int}} = -\sigma_n \log P^E_t + \beta_X \log P^X_{int} + h_{2,n}(t) + \lambda_{2,in} + \varepsilon_{2,int}, \tag{19}$$

where $h_{2,n}(t)$ is a sector-specific flexible time trend, $\lambda_{2,in}$ is a firm fixed effect, and $\varepsilon_{2,int}$ is the error term. We identify σ_n from the coefficient $-\sigma_n$ on the log emissions price. When permit prices rise, firms reduce emissions relative to the composite, and the magnitude of this response reveals σ_n.[24] The firm fixed effect absorbs the time-invariant share parameter and baseline emissions terms, and we include $\log P^X_{int}$ as a control with coefficient β_X that we do not interpret causally. We estimate this regression separately for each sector n and cluster standard errors by firm and year.[25] Our baseline specification uses a linear sector-specific trend.

Similar to the inner nest, the price of emissions is endogenous. As an example, when firms improve their clean technology, their production processes become less emissions-intensive, reducing aggregate permit demand and driving down equilibrium permit prices. The sector-specific time trends $h_{2,n}(t)$ absorb smooth technological progress, but concerns remain if technology and prices comove at higher frequencies. Identification is further complicated because all firms face a common nominal emissions price.

We instrument for the emissions price using the carbon price surprise series described in Section 2.2, aggregated to an annual frequency by summing within each calendar year. These high-frequency price movements around discrete regulatory announcements are allowance supply shocks, purged of slow-moving macroeconomic, financial, and energy market conditions, and plausibly uncorrelated with firms' technological choices, which evolve over longer horizons.[26]

[24] Under CES, the coefficient on $\log P^X$ could also identify σ_n. We focus on $\log P^E$ because our identification strategy instruments for the emissions permit price using the carbon price surprise series.

[25] Because X_{int} and P^X_{int} depend on the first-stage estimate $\hat{\xi}_n$, standard errors come from a two-stage pairs cluster bootstrap that reestimates ξ_n in each replicate and propagates that uncertainty to σ_n. The firm and year variances are combined by two-way inclusion-exclusion (Cameron et al., 2011), with each bootstrap variance estimated from the interquartile range of the replicates, rescaled by the interquartile range of a standard normal, rather than from their standard deviation. The just-identified IV estimator has no finite moments (Kinal, 1980), so occasional extreme replicates—most likely when resampling the 17 year clusters—would dominate the standard deviation, while the quantiles of the replicate distribution remain well behaved. Appendix Figure A3 shows that inference is robust to a variety of other approaches. For the residual other non-manufacturing sector, we substitute parametric $\hat{\xi}_n$ draws from its asymptotic normal with dimension-matched cluster-robust standard errors, as the inner nest IV was numerically unstable on a subset of resamples.

[26] Other emissions-related policies—such as renewable portfolio standard-style policies—do not threaten identification of σ_n as long as they are not time-varying or their time-varying component is unrelated to the allowance supply

Clean technology is what this regression leaves unexplained: the firm fixed effect, the sector trend, and the residual together capture the part of a firm's emissions intensity that the permit and composite prices do not account for, which equation (5) identifies with $(\sigma_n - 1)\log B_{int}$. Appendix D shows how to invert the regression equation to recover B_{int} at the firm-year level.

Validation via Morishima Elasticities The nested CES restricts emissions to be equally substitutable with capital, labor, and materials. We test this assumption in Appendix A.1 by estimating pairwise Morishima elasticities between emissions and each productive input. Under nested CES all three should equal σ_n. We estimate Morishima elasticities that are uniformly small in magnitude and statistically indistinguishable from zero and across input pairs. An estimation approach allowing for a variable elasticity indicates they are relatively constant over the range of prices in the sample.

4.2 Reallocation Parameters

Carbon pricing raises unit costs more at dirty firms than at clean ones, shifting output shares across firms and across sectors. Three objects govern the aggregate elasticity of substitution summarizing this margin (Section 3.3): the emissions heterogeneity index χ_{nt}, the within-sector demand elasticity ϵ_n, and the cross-sector demand elasticity υ. We recover each from the data in turn.

4.2.1 Heterogeneity Index χ_{nt}

We use equation (12) to calculate χ_{nt} for each sector-year directly from the data, evaluating each firm's emissions cost share $s_{E,int}$ at the observed permit price and the composite prices P^X_{int}. Confidence intervals come from resampling firms within sector.

4.2.2 Demand Elasticities ϵ_n and υ

Our model's demand system (Section 3.2.2) is CES at both the firm and sector levels, with monopolistically competitive firms setting prices as a constant markup over marginal cost. The markup identity $\mu_n = \epsilon_n/(\epsilon_n - 1)$ in equation (7) lets us recover the within-sector demand elasticity ϵ_n from observed markups. We compute the sector-level markup μ_n by pooling revenue and variable costs across firms and years within each sector:

$$\mu_n = \frac{\sum_t \sum_{i \in \mathcal{I}_n} R_{int}}{\sum_t \sum_{i \in \mathcal{I}_n} \left(P^E_t E_{int} + P^X_{int} X_{int} \right)},$$

where R_{int} is firm i's revenue. Given the CES structure, we then back out the demand elasticity as $\epsilon_n = \mu_n/(\mu_n - 1)$.

We estimate the cross-sector demand elasticity υ from final demand, using nominal final demand expenditure by product from the Eurostat use tables and sector price indices from EU KLEMS

shocks in our instrument.

(Section 2.2).[27] Taking logs of the resulting demand curve gives our estimating equation:

$$\log V^F_{nct} = (1 - \upsilon) \log P_{nct} + \alpha_n + \alpha_{ct} + \varepsilon_{3,nct}, \tag{20}$$

where V^F_{nct} is nominal final demand expenditure on sector n in country c and year t, and P_{nct} is the sector's price index. We use expenditures instead of quantities because quantities are not reported in the data. To identify υ we instrument $\log P_{nct}$ with the intermediate input price index. Shifts in this index move firms' materials costs and hence the prices firms charge households. Since households do not purchase intermediate inputs directly, the index affects final demand only through P_{nct}.

4.3 Clean Technology Dynamics

We take the dynamic extension of Section 3.4 to data by estimating equation (17) horizon by horizon using local projections. In our implementation, we replace clean technology with the observable composite-to-emissions ratio. In the case where $\sigma_n = 0$ as we will estimate below, using this observable ratio as the outcome rather than clean technology generates the same estimates, which tell us how much firms' clean technology adjusts at horizon h in response to a carbon price shock at $t-1$ (see Appendix I). The estimating equation is:

$$\log \frac{X_{in,t+h}}{E_{in,t+h}} - \log \frac{X_{in,t-1}}{E_{in,t-1}} = \gamma^E_{h,n} \log P^E_{t-1} + \gamma^X_{h,n} \log P^X_{in,t-1} + \alpha^{(h)}_{in} + \eta_{in,t+h}, \qquad h \in \{0,1,2,3,4\}. \tag{21}$$

The left-hand side cumulates changes between $t-1$ and $t+h$, so the coefficient at $h=4$ is a five-year cumulative elasticity. The dating of prices at $t-1$ matches the time-to-build structure in equation (14). Prices at $t-1$ shift clean technology investment $I^B_{in,t-1}$, which raises clean technology beginning in period t.

We estimate equation (21) separately by sector and compute the impulse response functions out to five years. Since the permit price is endogenous, we instrument for $\log P^E_{t-1}$ using the lagged carbon price surprise series as we did when estimating σ_n. All specifications include firm fixed effects. For inference, we use Driscoll-Kraay standard errors, which allow for serial correlation induced by the cumulative horizon-h outcome and for cross-sectional dependence. Appendix I shows that, under the implementation lag and no-depreciation assumptions, the local projection specification recovers a dynamic investment policy response to a change in the carbon price.

[27] Three KLEMS industry groups each span two of our model sectors, so we merge those sector pairs and estimate on the resulting eight sector groups. Under CES demand with a common υ, expenditure on any union of sectors follows the same demand curve with the union's CES price index, so the group-level regression identifies the same υ. We exclude public administration, education, and health (NACE sections O, P, and Q), whose output is largely non-market and valued at cost.

5 Results

This section presents the estimates of the key model parameters, organized around the three margins developed in Section 3. Section 5.1 reports the firm-level production elasticities that identify the short-run input substitution margin. Section 5.2 reports the two ingredients that govern the reallocation margin: the emissions heterogeneity index χ_{nt} and the within- and cross-sector demand elasticities. Section 5.3 reports our local projection estimates of the long-run clean technology response to carbon pricing. Section 5.4 synthesizes these estimates into marginal abatement cost curves at the sector and economy-wide levels. Section 5.5 reports robustness checks.

5.1 Production Elasticities

Our nested CES production structure requires estimating two elasticities: the inner nest elasticity ξ_n governing substitution among capital, labor, and materials, and the outer nest elasticity σ_n governing substitution between the productive composite and emissions.

5.1.1 Inner Nest: Capital-Labor-Materials (ξ_n)

Figure A4 in Appendix A presents sector-specific estimates of ξ_n from equation (18). Estimates for most sectors lie near one, and for most sectors we cannot reject Cobb-Douglas ($\xi_n = 1$): point estimates lie modestly above one in some sectors (e.g., rubber and plastics) and below one in others (e.g., cement), but these differences are imprecisely estimated. The power and mining sectors have imprecise negative point estimates—neither rejects Cobb-Douglas—which we replace with the smallest positive sector estimate in quantification (Appendix A).[28] Appendix Figure A5 shows that these estimates are stable across different ways of constructing the wage GIV.

The inner nest elasticities affect our analysis through their role in constructing the composite input price P^X_{int} and quantity X_{int}. The variation in ξ_n across sectors implies different aggregation weights when combining capital, labor, and materials into the productive composite and shapes how firms respond to materials price fluctuations induced by permit price shocks.

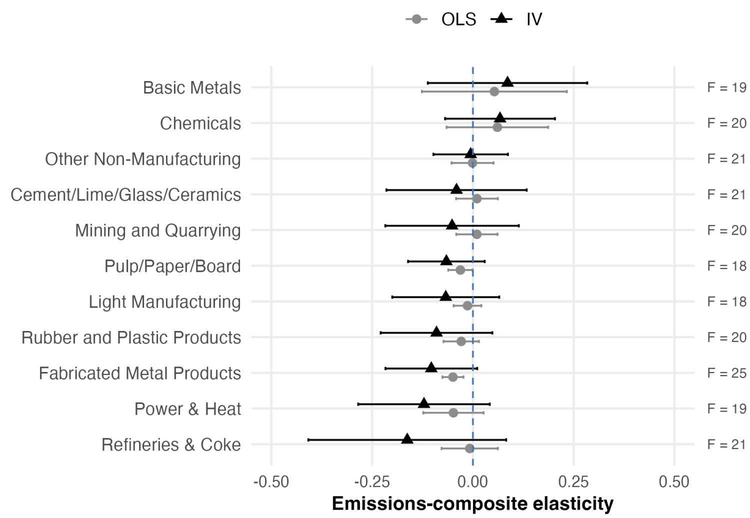
5.1.2 Outer Nest: Emissions-Composite (σ_n)

The central parameter governing firm-level marginal abatement costs is the elasticity of substitution σ_n between emissions and the productive composite. Figure 2 presents our sector-specific estimates from equation (19), with OLS estimates in gray and IV estimates in black.

All estimates cluster near zero, with OLS and IV point estimates similar across all sectors. We reject Cobb-Douglas ($\sigma_n = 1$) for every sector, with all estimates economically close to, and statistically indistinguishable from, the Leontief case ($\sigma_n = 0$). The IV first stage is relatively strong: Kleibergen-Paap first-stage F statistics, reported in the right margin of Figure 2, exceed

[28] Our main counterfactual results are robust to this imputation choice; for example, replacing both estimates with the median positive sector estimate instead leaves the aggregate counterfactual results essentially unchanged and preserves qualitative sector-level patterns.

Figure 2: Emissions-Composite Elasticity (σ_n)



Notes: This figure plots sector-specific estimates of the elasticity of substitution σ_n between emissions and the productive composite from equation (19). Gray circles show OLS estimates; black triangles show IV estimates using the carbon price surprise series. Error bars show 95% confidence intervals based on a two-stage cluster bootstrap by firm and year (see footnote 25 in Section 4.1.2). The dashed vertical line marks zero (Leontief). The right margin reports each sector's Kleibergen-Paap first-stage F statistic for the IV estimate.

17.7 in every sector. These uniform near-zero elasticities imply that firms have extremely limited ability to substitute away from emissions toward other inputs. The only way for firms to reduce emissions in the short run is to cut output.

The technological rigidity of the emissions input mix contrasts sharply with that of labor, capital, and materials in the inner nest. Firms have almost no ability to trade off the productive composite against emissions because emissions embodied in production are determined by the technology of the equipment already installed.

The finding that σ_n is approximately zero may appear counterintuitive for sectors like power, where the documented transition from coal to natural gas and renewables suggests substantial scope for input switching. The median firm in our panel operates only one facility (Figure A6), so the within-firm input substitution margin is mechanically narrow. Cost-weighting the regression does not reveal a positive elasticity masked at large firms (Appendix Table A2). In Appendix Section G we present case studies of the largest EU-ETS power firms to show why. These firms run baseload-dominant fleets, and as EU-ETS prices rose sharply over 2015–2021, coal capacity factors fell sharply at the firms with sizable coal and gas fleets while gas capacity factors at the same firms did *not* rise correspondingly. Coal-dominant firms, with no gas to switch to, kept running coal. In firms with matched generation data we see no strong evidence of within-firm input substitution. Baseload and flexible units are not substitutes within the firm even when the carbon price rises. The displaced coal generation flowed to renewables and imports at the system level, consistent with our overall finding that short-run emissions reductions in this setting operate primarily through across-firm reallocation rather than within-firm input substitution.

A further concern is that our outer nest elasticity estimates do not account for entry and exit of firms. If variation in carbon prices leads firms with different emissions intensities to enter or exit, our estimate—based purely on within-firm variation—might miss this margin of adjustment. Appendix Figure A7 presents an alternative set of outer nest elasticity estimates utilizing a random effects (at the firm level) estimator, which combines the within-firm variation used in our baseline with the between-firm variation from sources such as entry and exit driven compositional changes over time. The results suggest that accounting for entry and exit does not meaningfully affect the magnitude of the outer nest elasticity in our panel of EU firms.[29]

5.2 Reallocation Parameters

We now turn to the reallocation margin, which operates across firms and sectors. The elasticity of sector-level marginal abatement curves depends not only on firm-level substitution elasticities but also on the scope for reallocation across heterogeneous firms. Two parameters govern the within-sector reallocation channel: the heterogeneity index χ_{nt}, which captures within-sector dispersion in emissions cost shares across firms, driven to a first order by clean technology, and the within-sector demand elasticity ϵ_n, which determines how readily consumers shift toward cleaner producers when carbon pricing raises dirty firms' costs. A third parameter, the cross-sector demand elasticity υ, governs reallocation across sectors and the economy-wide elasticity. Figure 3 presents estimates of χ_{nt}, ϵ_n, and the sector-level elasticities of substitution; we report υ separately below.

5.2.1 Emissions Heterogeneity Indices χ_{nt}

Figure 3a reveals substantial variation in emissions heterogeneity across sectors. The power sector exhibits the highest heterogeneity index at approximately 0.10, reflecting the wide dispersion in clean technology and emissions intensity across firms. At the other end, paper has an index of 0.01, indicating more homogeneous emissions profiles.[30] The magnitude of these heterogeneity indices matters for the aggregate elasticity. Since our estimated $\hat{\sigma}_n \approx 0$ for all sectors, the within-firm term vanishes and the sector-level elasticity is $\bar{\sigma}_{nt} = \chi_{nt}\epsilon_n$, adjusted in our quantification by the pass-through correction of Appendix H.2.

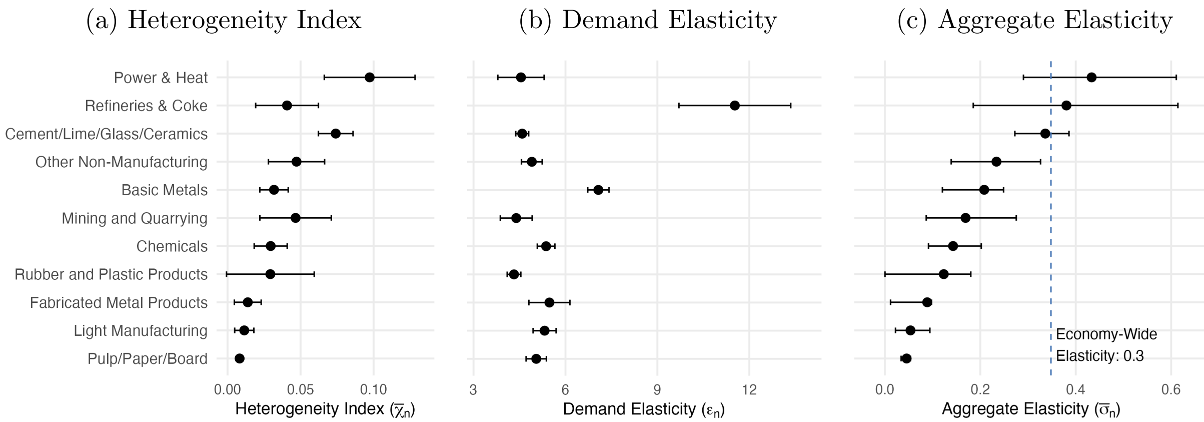
5.2.2 Within-Sector Demand Elasticities ϵ_n

Figure 3b reports the estimated demand elasticities, which vary by a factor of three across sectors. Refining exhibits the highest demand elasticity at 12, reflecting the commodity nature of refined fuels and consumers' ability to substitute across suppliers. Basic metals also shows a higher demand elasticity of 7. At the other extreme, rubber and plastics has a demand elasticity around 4,

[29] An important caveat is that Orbis has been shown to be limited in its ability to recover aggregate patterns related to entry and exit (Bajgar et al., 2020; Kalemli-Özcan et al., 2024). So, while the estimates presented in Appendix Figure A7 suggest our results are robust to entry-exit patterns observed within Orbis, it is likely Orbis is not well suited to definitively capture the degree to which this channel can explain aggregate trends.

[30] Appendix Figure A11 shows the underlying within-sector distribution of emissions cost shares, the key ingredient for the heterogeneity index, for the largest emitting sectors.

Figure 3: Heterogeneity Index, Demand Elasticities, and Aggregate Elasticities



Notes: Panel (a) shows the heterogeneity index χ_{nt} computed within each sector-year following equation (12). The plotted value is the average of χ_{nt} across sample years, denoted $\bar{\chi}_n$. Panel (b) shows sector demand elasticities ϵ_n estimated from observed markups: $\epsilon_n = \mu_n/(\mu_n - 1)$, where μ_n is the sector-level markup pooled across firms and years. Panel (c) shows the aggregate elasticity $\bar{\sigma}_n$ in the Leontief case, computed from the time-averaged heterogeneity index and demand elasticity using the exact expression derived in Appendix H.2. Error bars show 95% confidence intervals: panel (a) resamples firms within sector (999 replications), panel (b) uses the delta method, and panel (c) pairs each joint resample of the heterogeneity index and pass-through correction with a demand elasticity drawn from its asymptotic normal, reporting percentile intervals. The dashed line corresponds to the aggregate, economy-wide elasticity derived in Appendix H.3.

and power has a demand elasticity around 5, consistent with more differentiated products or less consumer flexibility.

5.2.3 Aggregate Elasticities by Sector $\bar{\sigma}_n$

Figure 3c shows the estimated aggregate elasticities implied by the demand elasticities and the heterogeneity index when we adopt the Leontief firm elasticity ($\sigma_n = 0$) found in Section 5.1.2. The top polluting sectors like power, refining, and basic metals have the most elastic response to emissions pricing. These sectors have substantial cross-firm heterogeneity in emissions intensity—especially power and basic metals—while refining and basic metals exhibit highly elastic demand.[31]

5.2.4 Cross-Sector Demand Elasticity υ

We also estimate the elasticity of substitution across sectors, υ, which governs reallocation of final demand from emissions-intensive to cleaner sectors. Table A4 in Appendix A reports estimates of equation (20). The OLS price coefficients in columns (1)–(3) exceed one, implying a negative υ: demand shocks raise expenditure and prices together, biasing the coefficient upward.

To address this simultaneity, columns (4)–(6) instrument the price with the intermediate input price index, which yields a strong first stage. Our preferred specification is column (6), which uses total final use to measure final demand. Household and total final consumption in columns

[31] The confidence interval for fabricated metal products is asymmetric, with the point estimate near its upper limit. Dispersion in this sector comes from a small number of emissions-intensive firms whose emissions cost shares lie far above the sector median.

(4)–(5) give similar point estimates with wider confidence intervals. We estimate $\upsilon \approx 1.1$, close to Cobb-Douglas demand. Expenditure shares respond only mildly to relative prices and across-sector reallocation provides a modest additional margin for emissions reductions beyond within-sector reallocation. The aggregate economy-wide emissions elasticity using this demand elasticity estimate is plotted as the dashed line in Figure 3c. Its value is 0.3, comparable to the most elastic sectors.

5.3 Clean Technology Dynamics

Figure 4 plots our estimates of the five-year cumulative effect of a permit price shock on clean technology, $\hat{\gamma}^E_{4,n}$ from equation (21). The IV estimates span a modest range, approaching 0.6 for the most responsive sectors. Refining has the largest five-year point estimate ($\hat{\gamma}^E_4 = 0.564$, SE 0.148), followed by power (0.425, SE 0.123) and mining and quarrying (0.256, SE 0.152). A positive elasticity means that a higher permit price raises the composite-to-emissions ratio: holding productive inputs fixed, these firms achieve lower direct emissions per unit of input, consistent with the directed technical change prediction that firms exposed to the carbon price invest in cleaner production. Because the carbon price surprise series isolates unanticipated movements in the permit price that are also likely perceived as persistent, we read these local projections as an approximation to the dynamic response of clean technology investment to a carbon price change.[32]

The remaining sectors have smaller positive point estimates and fabricated metals is the only sector with a small negative point estimate ($\hat{\gamma}^E_4 = -0.044$, SE 0.087), but it is statistically indistinguishable from zero. Chemicals (0.011, SE 0.064) and basic metals (0.033, SE 0.157) have essentially zero point estimates.[33] Appendix Figure A9 shows the full sector-level impulse response functions.

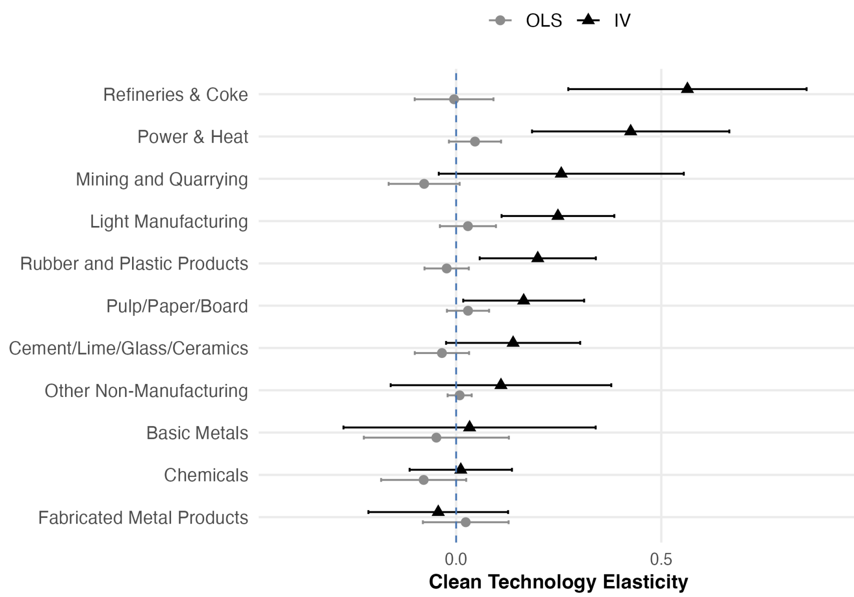
Estimating a pooled five-year elasticity yields $\hat{\gamma}^E_4 = 0.211$ (SE 0.043) when firms are weighted equally, and 0.318 (SE 0.055)—about 50% larger—when weighting observations by its sector's 2005 verified emissions share. The larger emissions-weighted response shows that carbon-intensive sectors—which account for the bulk of aggregate emissions—invest in clean technology more than the average firm. Appendix Figure A10 plots both pooled responses by horizon. In both, improvements accrue incrementally over the five years following the shock rather than on impact.

External Validation Against Colmer et al. (2024) Colmer et al. (2024) provide a useful benchmark for our estimates. Using a matched difference-in-differences design on a subset of French manufacturing, they estimate a treatment effect of -0.174 on logged emissions intensity during EU-ETS Phase II. Re-estimating our local projection on the corresponding manufacturing

[32] Permit banking links current spot prices to future permit prices so that unanticipated price shocks signal persistent changes.

[33] Investment that makes the composite input more productive, such as materials-saving efficiency improvements, would register as a *decline* in measured clean technology. In the presence of input-output linkages, a carbon price also incentivizes these kinds of investments. If this occurs in the data, our estimates understate the clean technology response.

Figure 4: EU-ETS Price Elasticity of Clean Technology



Notes: This figure plots sector-specific estimates of the five-year cumulative coefficient $\gamma^E_{4,n}$ from equation (21), whose outcome uses the composite quantity $X_{int} = (C^K_{int} + C^L_{int} + C^M_{int})/P^X_{int}$. Gray circles show OLS estimates and black triangles show IV estimates. Each estimate comes from a separate sector-level local projection with firm fixed effects, and the baseline specification includes the $\log P^X_{in,t-1}$ control. The $t-1$ permit price is instrumented with the lagged carbon price surprise series. Error bars show 95% confidence intervals based on Driscoll-Kraay standard errors. The dashed vertical line marks zero.

sectors yields $\hat{\gamma}^E_4 = 0.119$ (standard error 0.067).[34] Because Colmer et al. (2024) estimate the treatment effect of introducing the ETS whereas we estimate a price elasticity, we need a baseline non-ETS carbon price faced by comparison firms to translate their effect into price units. Using baseline prices between 2 and 6.5 EUR/tCO_2—a range supported by French statutory fuel excises and OECD effective carbon-rate estimates—their treatment effect implies an elasticity of 0.086–0.157, which brackets our estimate (see Appendix J for details). Thus, despite using different samples and identification strategies, the two approaches imply similar longer-run responses to carbon prices by firms.

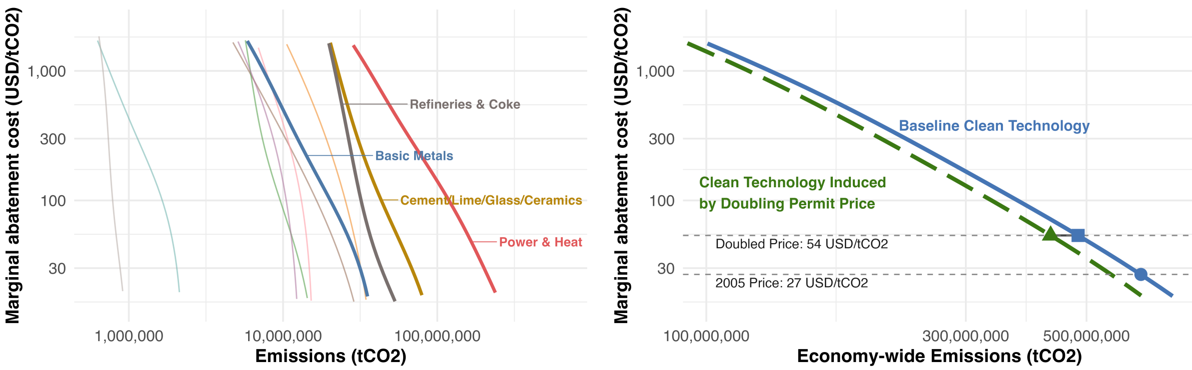
5.4 Marginal Abatement Cost Curves

We now construct marginal abatement cost (MAC) curves implied by our model. Figure 5 presents sector-level and economy-wide MAC curves on a log-log scale, averaged across the sample years as baseline-revenue-weighted geometric means. Each slope is inversely proportional to the corresponding aggregate elasticity as discussed in Section 3.3.

Panel (a) plots the sector MAC curves, with the four largest-emitting sectors highlighted: each

[34]The overlapping sectors are basic metals; cement, lime, glass, and ceramics; chemicals; fabricated metal products; pulp, paper, and board; and rubber and plastic products. We also estimate a version including light manufacturing, which partially overlaps.

Figure 5: Sector Marginal Abatement Costs and the Role of Clean Technology



(a) Sector MAC Curves (b) Economy-Wide MAC: Clean Technology

Notes: Panel (a) plots the sector aggregate MAC curves. The four largest-emitting sectors are highlighted and labeled, and the remaining sectors appear as thin unlabeled curves. Panel (b) plots the economy-wide aggregate MAC under baseline clean technology, the level prevailing in each sample year (solid blue), and under the advanced clean technology level induced by a doubling of the permit price (dashed green). The gray dashed horizontal lines mark the average 2005 permit price and twice that level. Both curves are marked at the doubled price, and the baseline-technology curve also at the 2005 price. All curves are displayed in log-log space. The sector and economy-wide aggregates are constructed by simulating our model on a grid of counterfactual permit prices in each year of 2005–2021, recovering the corresponding sector and economy-wide aggregate emissions, averaged across the sample years as baseline-revenue-weighted geometric means. Following the definition of the MAC, the relevant aggregate is held fixed along each curve: sector output for the sector curves and aggregate household final demand for the economy-wide curves.

traces how sector emissions fall as the carbon price rises, holding sector output fixed. Power—the largest emitter—also has the flattest curve, while the curves of several smaller sectors are close to vertical. The easiest abatement is concentrated precisely where emissions are largest. The median sector has an elasticity of about 0.2: by shifting output from dirtier to cleaner producers, a sector abates more cheaply than any of its constituent firms.

Aggregating one step further, the blue curve in Panel (b) plots the resulting economy-wide MAC at the baseline clean technology level prevailing over the sample years. It is flatter still than the median sector curve. Appendix Figure A12 isolates each margin's contribution and shows that adding reallocation across sectors flattens the curve beyond what reallocation within sectors already delivers, but within-sector reallocation is the primary driver for why the economy-wide MAC is relatively flat.

Panel (b) also shows how improvements in clean technology shift the MAC downward. The dashed green curve traces the MAC schedule with clean technology improvements caused by a doubling of the carbon price, recovered using the five-year local projections estimate. Moving from the solid blue curve to the dashed green one lowers the MAC by about 14 USD/tCO_2 at the doubled price. Increasing the carbon price to twice its 2005 level delivers 53.0 million tons of additional abatement in total—36% more than the price increase would achieve by moving along the solid blue curve, holding clean technology fixed.[35]

[35]The elasticity of the economy-wide MAC is similar across the two curves since clean technology primarily shifts

5.5 Robustness

Appendix A collects robustness and specification checks for our two central estimation results: the near-zero outer nest elasticity σ_n and the clean technology response to carbon pricing. For the outer nest elasticity, the Morishima elasticities between emissions and each productive input (Section 4.1.2) are statistically indistinguishable from one another and uniformly close to zero, supporting the nested CES restriction that emissions are equally substitutable with capital, labor, and materials (Appendix Tables A10 and A11). The finding that $\sigma_n \approx 0$ is also robust to quadratic, cubic, and natural spline time trends, including under inference that remains valid when instruments are weak (Appendix Figures A13 and A15); to instrumenting with the carbon price surprise series and the wage GIV together (Appendix Figure A14); to the 2013 expansion of EU-ETS coverage under Phase III (Appendix Table A5); and to weighting firms by total input cost following Oberfield and Raval (2021) (Appendix Table A2), which provides no evidence that large firms have a meaningfully positive substitution elasticity. Nor is inference fragile to the small number of year clusters ($G = 17$): wild cluster bootstraps on the year dimension (Cameron et al., 2008; MacKinnon et al., 2021) yield similar confidence intervals (Appendix Figure A3). For the clean technology response, the estimated impulse response accumulates roughly linearly across horizons rather than arriving on impact or decaying, consistent with the one-period implementation lag and no-depreciation assumptions behind our law of motion (Appendix I). Inference for the local projections is likewise robust to wild cluster bootstrapping to handle the small number of year clusters (Appendix Figures A9 and A10).

6 Counterfactuals

We now use the estimated model to run counterfactuals that isolate the equilibrium effect of changes in the EU-ETS permit price. Our central counterfactual doubles the EU-ETS permit price—approximately its observed 2005–2021 increase—and traces the effect through the full production network. We distinguish a short run, in which clean technology is fixed, from a long run, in which firms' clean technology investment responds over the following five years. We simulate this price shock in every sample year and report the baseline-revenue-weighted geometric mean effect across years.

6.1 Aggregate Results

Table 1 quantifies the effects of doubling the carbon price. The top row shows that, in the short run with clean technology fixed, doubling the carbon price cuts emissions by 23%. A higher cost of emissions results in output falling 3.6% and a corresponding increase in output prices of 2.8%. Since emissions and labor are complements, employment falls 1.0%.[36]

the level of the curve rather than the slope.

[36] For a marginal, 1% price change, Appendix Table A6 reports the corresponding elasticities: the emissions elasticity (-0.34) is an order of magnitude larger in absolute value than the output (0.04) and employment (0.01) elasticities.

Table 1: Equilibrium Effects of Doubling the Carbon Price

	Δ Emissions (%)		Δ Output (%)		Δ Prices (%)		Δ Labor (%)	
	Total	IO share	Total	IO share	Total	IO share	Total	IO share
Reallocation: Active								
Short run	-22.89%	0.07	-3.62%	0.83	2.80%	0.75	-0.98%	0.35
Long run	-31.15%	0.03	-2.51%	0.83	1.87%	0.74	-0.67%	0.41
Reallocation: Suppressed								
Short run	-4.70%	0.82	-3.87%	0.79	3.24%	0.74	-2.25%	0.63
Long run	-20.63%	0.12	-2.50%	0.78	2.07%	0.74	-1.49%	0.64

Notes: This table reports counterfactual outcomes for ETS-regulated firms from 2005 to 2021 under a doubling of the carbon price (the approximate 2005–2021 change). Each year's equilibrium uses the firms observed in that year, and annual effects are baseline-revenue-weighted geometric means across years. *Reallocation: Active* allows within- and across-sector reallocation; *Reallocation: Suppressed* shuts down both margins. *Short run* holds clean technology fixed at baseline; *Long run* additionally incorporates the estimated, sectorally heterogeneous five-year change in clean technology. "IO share" is the share of each effect attributable to input-output network propagation, computed as the difference between the full-network equilibrium and an alternative equilibrium where the materials expenditure shares Ω_{mnt} in equation (8) are set to zero, expressed as a fraction of the total effect on the unit interval $[0, 1]$.

Over the long run, firms respond to the higher price by investing in clean technology. The second row includes the estimated five-year clean technology response and shows that emissions reductions deepen to 31%. Our counterfactual long-run prediction is broadly in line with the actual 35% decline in emissions at regulated installations over our sample, suggesting that carbon pricing was a central driver of the observed reduction. In addition to cutting emissions, clean technology eases other economic impacts in the long run. Output falls by only 2.5%, a 31% smaller fall than the short run, and the price and employment effects both shrink relative to the short run.

The lower two rows of Table 1 present results when suppressing reallocation by holding firms' relative output shares fixed. Without reallocation, emissions fall by 5% in the short run, roughly a fifth of what is achieved with reallocation. Since firm production is Leontief, this emissions reduction comes from output curtailment. Reallocation dampens the effects on consumer prices and quantities because it allows consumers to shift from relatively high price to low price producers. It also significantly blunts the aggregate negative effect on employment as cleaner firms shrink less or even grow compared to a world without reallocation. This finding has consequences for policy design. Carbon markets often include provisions like production-indexed rebates or exemptions for trade-exposed sectors to reduce cost impacts of carbon pricing. Our results suggest that policies that limit cost impacts will mute reallocation and blunt precisely the margin that delivers the short run abatement.

The input-output share columns in Table 1 split each effect into a direct effect of the carbon price and an indirect effect transmitted through the production network. The emissions reduction is almost entirely direct: network propagation contributes just 1.5pp of the 23% total. The indirect effect is small for emissions because the shock the network transmits is far smaller and lands on the wrong firms. Sector materials prices rise by only about 2.9% versus a doubling of the carbon price. Emissions intensive firms also devote less of their costs to materials, 57% against a sector average of 68%, with a within-sector correlation between carbon and materials cost shares of -0.45. The indirect effect through the input-output actually lowers the relative price of emissions-intensive firms by 0.3pp rather than raising it.

Unlike emissions, the economic impacts are mostly indirect. Network propagation accounts for most of the output decline (83%), the price increase (75%), and about a third (35%) of the employment effect. Network propagation scales up the common level of marginal costs, which moves output, prices, and employment, but it generates almost none of the within-sector relative price dispersion that reallocates market share from dirty to clean firms and drives the emissions decline.[37]

Appendix Table A7 presents results of the policy counterfactuals under different values of the demand elasticity, the production elasticities, and the choice of capital rental rate. The impacts on prices and output are relatively similar regardless of the parameter values. The impact on emissions, as one should expect, is sensitive to the demand elasticity and elasticity of substitution between emissions and the productive composite. More elastic demand and more elastic emissions substitution lead to greater emissions declines, as described above. The impact on labor is sensitive to the production elasticities. Greater ability to substitute in either nest leads to a smaller reduction in labor, or even an increase under some parameterizations.

6.2 Sector-Level Results

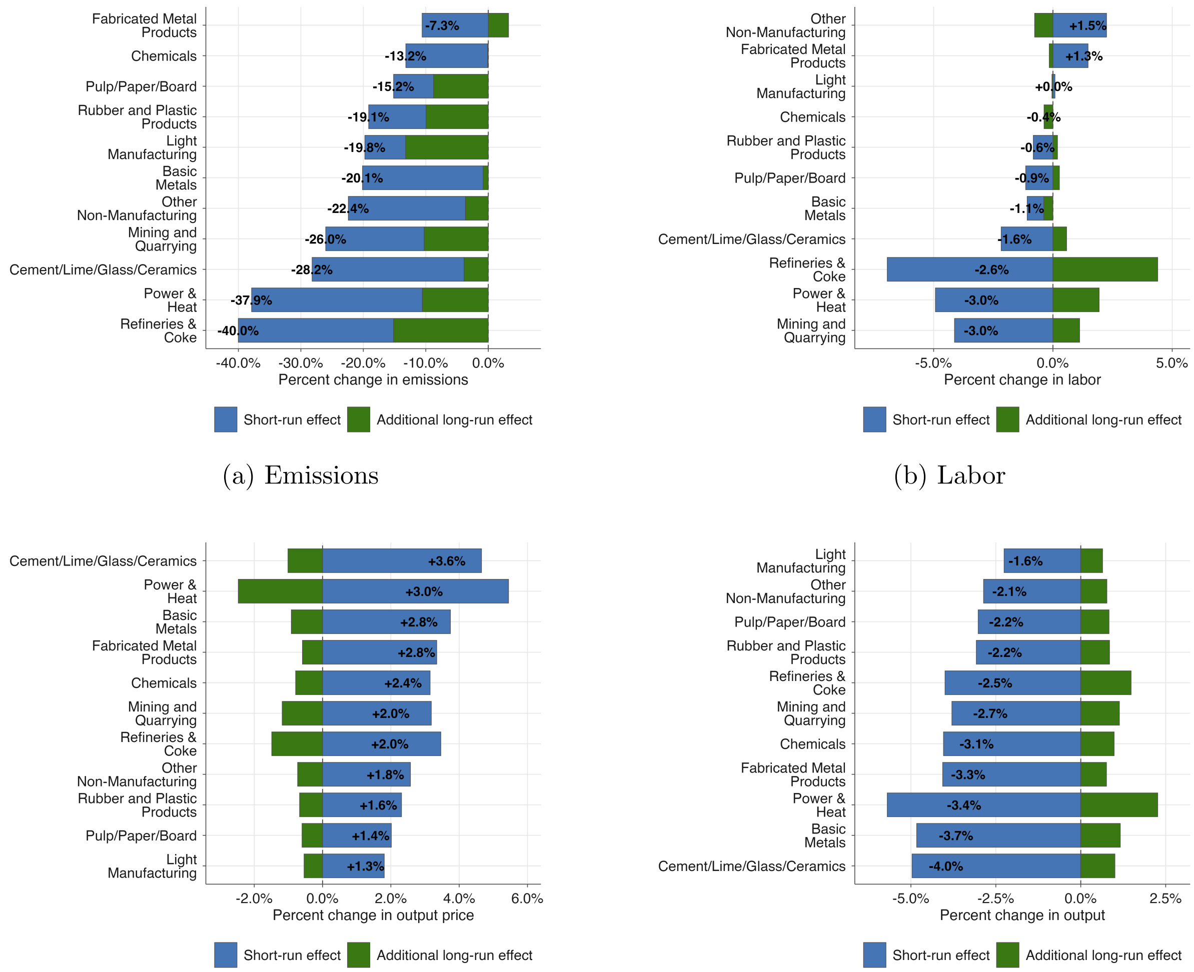
The aggregate effects mask wide variation across sectors. Figure 6 plots the sectoral impacts of doubling the carbon price, splitting each outcome into the short-run effect (blue) and the incremental five-year clean technology response (green). Their sum, labeled on each bar, is the long-run effect. Emissions fall the most in refining and in the power sector, which are the sectors with the greatest scope for reallocation given their elastic demand and within-sector dispersion in clean technology (Section 5.2.3). The clean technology contribution to emissions reduction is highly heterogeneous across sectors, reflecting the sector-level responses estimated in Section 5.3.

The price and output panels show where the economic impacts concentrate. Output prices rise in every sector, and by the most in the emissions-intensive ones such as cement, power, and basic metals, whose direct permit costs are largest and which bore the largest emissions declines. Output contracts most in these same sectors. Clean technology adoption blunts the price increases, again concentrated in power, and modestly cushions the output declines. Appendix Figure A16 decomposes the price changes and shows how the input-output network amplifies them.

Employment effects are the most uneven because of heterogeneity in the estimated inner nest elasticity ξ_n. Net job losses concentrate in the energy sectors—power, mining, and refining. Elsewhere, employment impacts are more mixed. As materials grow more expensive, firms substitute toward labor, so some sectors add workers even as their output falls. Clean technology adoption

[37] One notable result is that the input-output share of the price effect is very similar whether reallocation is active or suppressed. To a first-order, price pass-through is governed by the materials shares Ω_{mnt} and so is invariant to quantity reallocation.

Figure 6: Sectoral Short-Run Impacts and the Effect of Long-Run Clean Technology



(a) Emissions (b) Labor

(c) Output Prices (d) Output Quantities

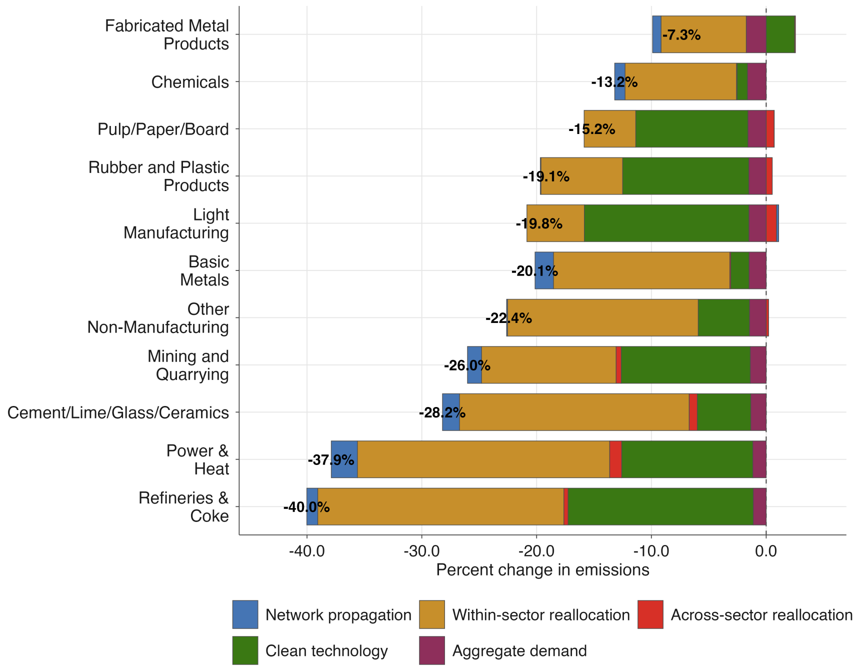
Notes: Panels (a)–(d) plot the change in emissions, labor employment, output prices, and output quantities by sector. The blue bar is the short-run effect of doubling the carbon price with clean technology fixed. The green bar is the additional effect of the estimated, sectorally heterogeneous five-year clean technology response, defined as the long-run effect minus the short-run effect. The sum is the long-run effect and is labeled on each bar. Annual sector changes are baseline-revenue-weighted geometric means across 2005–2021.

cushions the hardest-hit sectors, offsetting much of the short-run employment loss in refining and in power.

6.3 Decomposition: Role of Reallocation

Figure 7 decomposes each sector's emissions change into five channels, showing where the abatement originates. The channels correspond to a sequence of counterfactuals that switch on one adjustment margin at a time, so each bar is the incremental effect of a margin given the ones before it. Network propagation holds firm and sector market shares fixed and sector final demand quantities at baseline, so output moves only through demand transmitted along input-output linkages. The

Figure 7: Decomposition of Emissions Changes by Sector



Notes: This figure decomposes the change in emissions at ETS firms under a doubling of the carbon price and the associated estimated five-year change in clean technology. Network propagation captures the change in emissions from firms' output responses when firm and sector market shares remain fixed and sector final demand quantities are held at baseline levels, so demand shifts reach firms only through input-output linkages. Within-sector reallocation captures the additional change from shifting market shares across firms within sectors. Across-sector reallocation captures the additional change from shifting demand across sectors while holding aggregate final demand quantity fixed. Clean technology captures the additional change from the estimated sector-specific change in emissions productivity once all market shares adjust. Aggregate demand captures the additional emissions change when nominal spending is held fixed instead of real final demand quantity. Each cumulative effect is a baseline-revenue-weighted geometric mean across 2005–2021. The colored bars are differences between adjacent cumulative effects, are displayed in the order the effects are introduced, and sum exactly to the black percent change total.

two reallocation channels then let expenditure respond to the relative price changes induced by the carbon price, first across firms within sectors and next across sectors, holding the aggregate final demand quantity fixed. Clean technology then adds the estimated five-year change in clean technology with all market shares free to adjust, so it nets the direct decline in emissions per unit of output. Aggregate demand finally relaxes the fixed aggregate quantity, allowing demand to contract or grow at all firms.

Within-sector reallocation is the primary driver in most sectors and dominates in the highest-emitting sectors like refining, power, and cement. Across-sector reallocation is small everywhere: with $\upsilon \approx 1.1$, final demand is close to Cobb-Douglas, so relative price changes shift expenditure across sectors only weakly. Clean technology is one of the most heterogeneous channels, reflecting the sector-level responses estimated in Section 5.3. It has the largest impacts in refining and light manufacturing. Network propagation and aggregate demand contribute modest and fairly uniform emissions reductions. Network propagation is largest in sectors that sell a larger share of output as materials to other firms, while the uniformity of the aggregate demand channel is because it is a common proportional contraction from an increase in the aggregate price index.

6.4 Robustness: Carbon Leakage to Unregulated Firms

A natural concern with the EU-ETS is that carbon pricing shifts production from regulated to unregulated firms, undoing some of the intended emissions reductions (Section 3.5). To quantify the extent of leakage we use our extended model with the larger Orbis panel of non-ETS firms (Section 2.3). Because emissions at non-ETS firms are unobserved, we impute emissions intensities using several different strategies. Appendix Table A9 reports leakage rates. In the long-run counterfactual with the estimated clean technology response, leakage rates across the five strategies range from -0.7 to 2.2%. Leakage rates can be negative if emissions-intensive non-ETS firms shrink. This can happen if the materials cost increase from propagation through input-output linkages is sufficiently large for non-ETS emitters, or if the clean technology improvement at ETS firms leads to substantial reallocation toward ETS firms. Equilibrium leakage to unregulated firms is limited and does not materially change our main conclusions, consistent with prior evidence (Naegele and Zaklan, 2019; Colmer et al., 2024).

7 Conclusion

This paper develops a framework for quantifying the equilibrium effects of carbon pricing using commonly available data on firm financials and emissions. We isolate three adjustment margins: firms' short-run input substitution, reallocation of output from dirty toward cleaner firms, and firms' long-run adoption of clean technology. We show how to aggregate firm-level responses into equilibrium outcomes at the sector and economy-wide levels. Importantly, our approach is portable: it can be applied wherever pollution pricing programs generate emissions data that can be merged with firm financial information, providing a template for evaluating pollution pricing policies across regulatory contexts.

Applying the framework to EU-ETS, we find the input substitution margin is rigid: the firm-level elasticity of substitution between emissions and other inputs is near zero in every sector, so firms cannot reduce emissions in the short run without cutting output. Reallocation across firms is a major driver of the aggregate emissions response, with the biggest reductions concentrated in sectors with substantial cross-firm heterogeneity in clean technology, such as power and cement, and in sectors facing especially elastic demand, such as refining. Lastly, we show that clean technology adoption responds to carbon price shocks, with the largest clean technology responses in carbon-intensive sectors.

These findings help reconcile a puzzle: how have European emissions fallen so sharply while output and employment have remained relatively stable? The answer is that reallocation across firms and endogenous clean technology adoption are the dominant adjustment margins, while short-run within-firm substitution plays almost no role. Large aggregate emissions reductions coexist with moderate economic costs because market shares shift toward cleaner firms and because carbon pricing itself induces the technological progress that lowers emissions intensity over time. For policymakers, the implications are twofold. First, pollution pricing can deliver substantial emissions reductions without large economic disruptions, because abatement comes mainly from reallocation toward cleaner firms and from induced technology rather than from firms cutting output. Second, because clean technology takes years to build and pays off only if carbon prices stay high, the clean technology responses we estimate are consistent with firms investing when price increases are expected to persist. Credible, lasting policy is important for the long-run technology dividend.

Our main framework could be extended in several ways. Our headline counterfactuals are partial equilibrium, holding factor prices fixed, though we show in Appendix E that the approach can be easily extended to general equilibrium. The set of firms is fixed: entry, exit, and relocation abroad are absent, so our leakage estimates capture reallocation toward existing unregulated firms rather than the offshoring of production. Market structure is monopolistically competitive with constant markups, abstracting from strategic pricing responses to carbon costs and local concentration. And the clean technology response we estimate is a reduced-form deterministic investment margin of a set of potential pre-existing technologies rather than uncertain innovation.

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